IDEA for Teachers: Tutorial

Kolibri, Nextcloud, Wikipedia offline, Files Disk and Backup Disk

For Justice and the teachers he trains. Version 1.0, 13 October 2026

IDEA - offline school computing.
No real passwords are included. Demo passwords come from your IDEA contact.

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Part A: For teachers

1. What IDEA is

IDEA (Initiative for Digital Education in Africa) is a charity based in Belgium. It puts offline computer systems in rural schools. Everything works from a small computer in the school, without internet.

> **No internet needed.** Everything in this guide works inside the school. Nothing you do is sent to the internet.

1.1 Words used in this guide

Word	Meaning
IDEA box	The small Raspberry Pi computer in the school that runs the apps. (On the demo it is named `wizardly-hugle`; your school's box has its own name.)
Console	The web page that lists the apps and opens them.
IDEA shortcut	The one shortcut on a classroom computer's desktop. It finds the IDEA box and opens the Console (chapter 2).
System Disk	The IDEA box's own disk. The apps live on it.
IDEA Disk	A USB disk or stick that the Console has prepared. It can become a **Files Disk** or a **Backup Disk**.
Spare disk	An IDEA Disk that is still empty.
Files Disk	An IDEA Disk for your own files. It appears as the folder **School Files** in Nextcloud (chapter 9).
Backup Disk	An IDEA Disk that keeps copies of the apps' data (chapter 10).
USB stick	Your own USB stick (FAT32 or exFAT), filled on any computer. IDEA only reads it and never changes it (chapter 5).
Channel builder	The IDEA app that turns a folder of your videos and PDFs into a Kolibri channel (chapter 5).
Operator	The person at the school who looks after the IDEA box, with the IDEA contact on the phone (Part B).

1.2 The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (for example Khan Academy). You make classes, lessons and quizzes and see each learner's progress. With the **channel builder** you turn your own files into a Kolibri channel.	3, 4, 5
Nextcloud	Your own files: share worksheets with a class, write documents together in **Nextcloud Office**, collect homework.	6, 7, 9

App	What you use it for	Chapters
Kiwix	Wikipedia and other reference books, offline.	8

1.3 How to use this guide

- **Part A (chapters 1 to 10)** is for teachers.
- **Part B (chapters 11 and 12)** is for the school operator.
- Each chapter has numbered steps with screenshots. The screenshots come from the IDEA demo box and use the demo school **IDEA Demo School** with class **Grade 5** and the demo accounts in the appendix (`admin`, `teacher1`, `learner1` to `learner3`; each password is the account name followed by `911!`). Your school's names will be different.

2. Getting connected and opening an app

All apps open from one web page: the **Console**. How you get to the Console depends on the computer you use.

The school Wi-Fi networks

The IDEA router at your school makes two Wi-Fi networks. Their names and first passwords are the same at every IDEA school:

Network	Password (until your school changes it)	Who joins it	What it reaches
IDEA	`idea911!`	Students, and the classroom computers	Only the IDEA box: the Console and its apps. No internet.
IDEA-Teachers	`idea-teachers911!`	Teachers only	The IDEA box. Internet is **off** at first; your IDEA contact can switch it on when the head teacher asks.

- Give students only the **IDEA** password. Keep the **IDEA-Teachers** password among the teachers.
- **Change the passwords later.** These first passwords are easy on purpose, but anyone who knows them can join. Once IDEA is running at your school, the head teacher chooses new passwords and asks the IDEA contact to set them on the router. After that, every phone and laptop must join again with the new password.

2.1 On a classroom computer: the IDEA shortcut

The classroom computers (the student computers) are already connected to the school network. They have **one** shortcut on the desktop, named **IDEA**. There are no separate icons for Kolibri, Nextcloud or the other apps.

1. Double-click **IDEA** on the desktop. You can also find it in the app menu. It opens in Chrome.

> **Screenshot pending retake (IDEA-01):** Student computer desktop with the one IDEA shortcut.

2. A start page appears for a moment while it looks for the IDEA box. First it tries the IDEA box this computer belongs to, then the last IDEA box it opened, then any other IDEA box at the school. When it finds one, the Console opens by itself.

Screenshot pending retake (IDEA-02): The IDEA start page while it looks for the IDEA box (wording to follow the idea#202 start.html).

3. If the start page says **No IDEA server found**, check that the computer is connected to the school network. Then type the name on the box's sticker in the field on the start page, and press Enter. The sticker is hand-written and has two lines:

- line 1: the name, for example `brave-otter`
- line 2: the address, for example `http://brave-otter.local`

Type the first line, for example `brave-otter`: small letters, with `-` between the words. The start page then remembers that IDEA box.

Screenshot pending retake (IDEA-03): Start page: No IDEA server found, with the sticker-name field.

Two IDEA boxes at your school? The Console shows the apps of all of them. You can also open a specific box by typing its sticker name in the field on the start page.

2.2 On your own laptop, tablet or phone

1. Connect your device to the Wi-Fi: teachers join **IDEA-Teachers**, students join **IDEA** (see The school Wi-Fi networks above).
2. Open the browser and go to `http://idea.local`. The Console opens.
3. If it doesn't open, type the name on the box's sticker: go to the address on the sticker's second line, for example `http://brave-otter.local`.

Open the Console by hand

Use this on a device without the IDEA shortcut, for example your own laptop, when the steps above don't work.

1. Click in the browser's address bar and type `http://` followed by the name on the box's sticker and `.local`: `http://<sticker-name>.local`, for example `http://brave-otter.local`. Press Enter.
2. The Console opens.

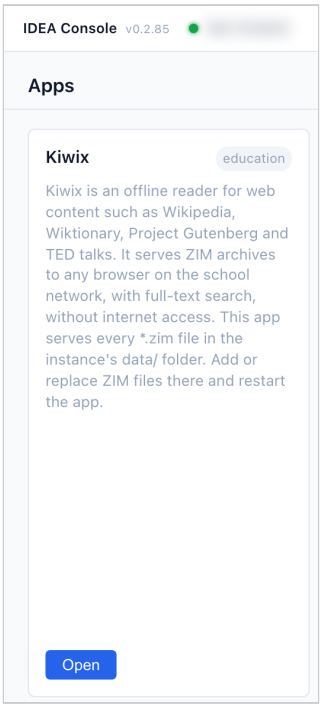


Figure 1. The Console opened by hand

3. **Make it your home page** (Chrome): open ⋮ **> Settings > On startup**, choose **Open a specific page or set of pages**, click **Add a new page** and paste the Console address.
4. **Bookmark it** (Chrome): press **Ctrl+D** (on a Mac ⌘**+D**), or click the star at the right of the address bar, then **Done**.

Steps 1, 3 and 4 take place in the browser's own window, so this tutorial has no screenshots of them. Other browsers have the same settings under similar names.

2.3 Open an app in the Console

1. The Console opens on the **Apps** page: one card per app, each with a short description and a blue **Open** button. The green dot at the top shows the Console is connected to the IDEA box.

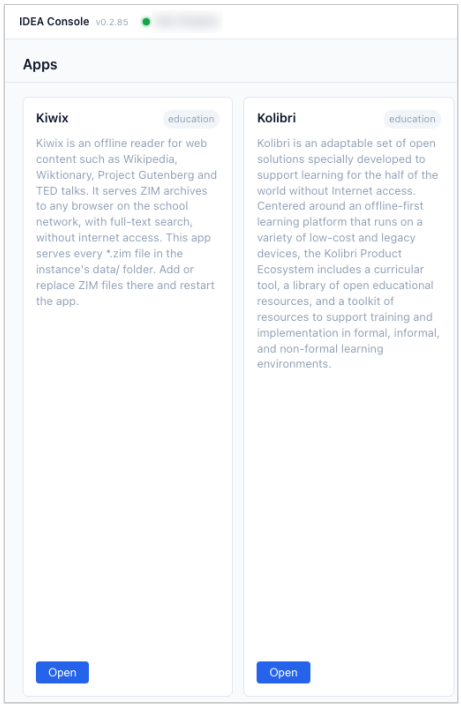


Figure 2. Console Apps page (two of the cards)

2. Click the blue **Open** button on the card of the app you need. The app opens in a new tab. Here it is Kolibri's sign-in page.

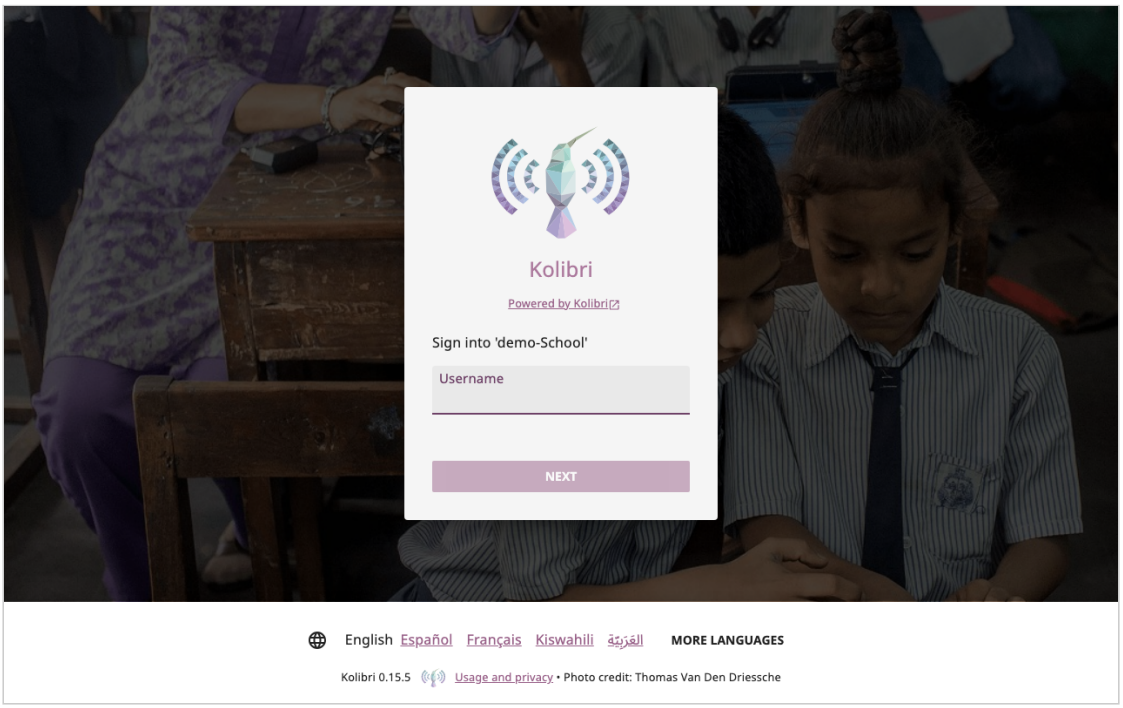


Figure 3. Kolibri opened from the Console

Good to know:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your school operator.
- Do not type numbers (addresses or ports) into the address bar. Always start from the Console and use **Open**.
- A student computer that no longer works properly is not repaired: the operator resets its card (chapter 13). Students lose only what they saved on the computer itself, so let them save their work in Nextcloud (chapter 7).
- The person icon at the top right (**Operator Login**) is for the operator (Part B). Teachers do not need it.

> **Tip.** You can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

Demo: school (facility) **IDEA Demo School**, teacher `teacher1` (password `teacher1911!`), class **Grade 5**. The demo lessons use the **Maths**, **Grade 5 Science** and **Early Reading** channels from the IDEA-DEMO disk; the quiz in 3.5 uses the exercises in **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions**.

3.1 Signing in

Open Kolibri from the Console (2.3). Type your username and click **NEXT**, then type your password.

3.2 Create a class

Creating classes needs a Kolibri admin account (in the demo school, `admin`, password `admin911!`; `teacher1` is the coach of Grade 5).

1. Open **Facility > CLASSES**. You see the list of classes.

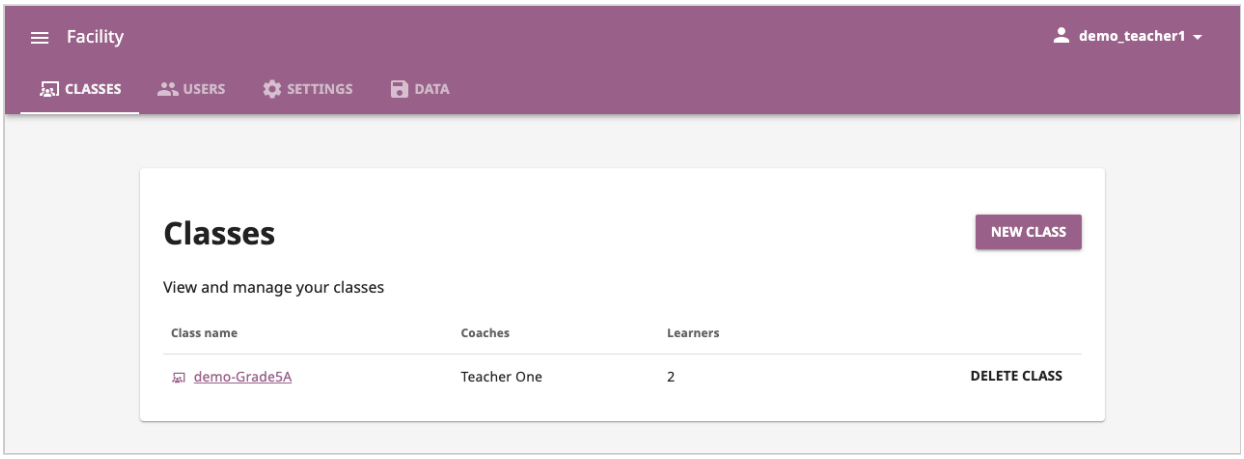


Figure 4. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed **CANCEL** because `Grade 5` already exists.)

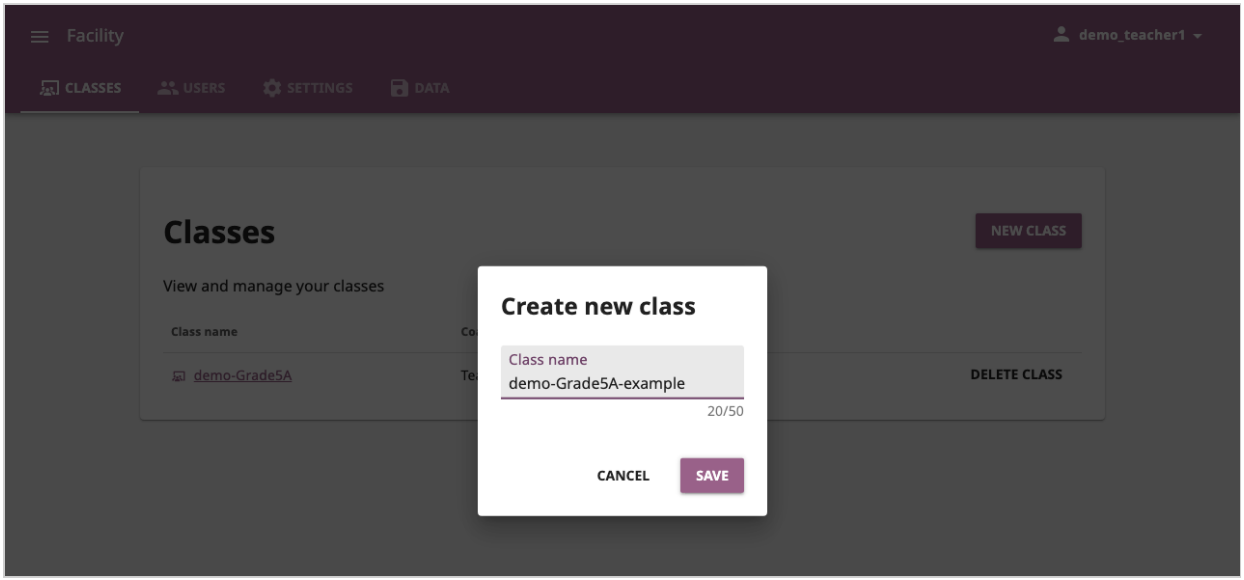


Figure 5. Create new class

3.3 Enroll learners

1. Click the class name. On the class page (with **Coaches** and **Learners**), click **ENROLL LEARNERS**. The **Enroll learners** page opens.

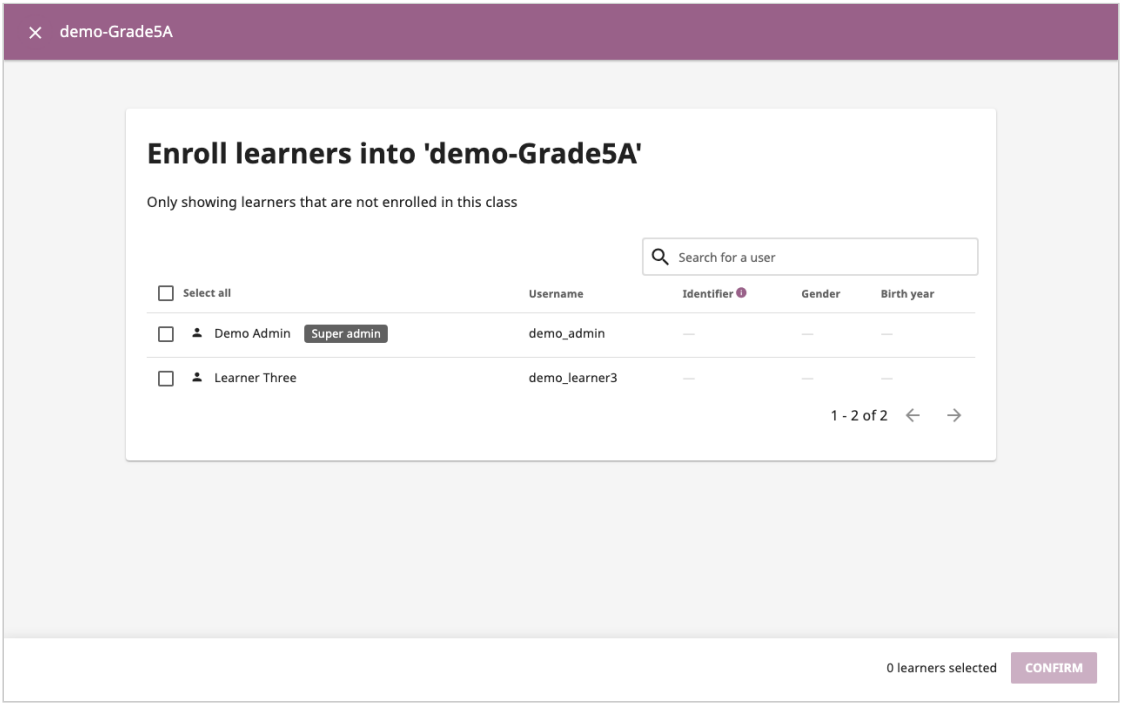


Figure 6. Enroll learners page

2. The list only shows learners who are not yet in the class. Tick each learner (here: Learner Three), then click **CONFIRM**.

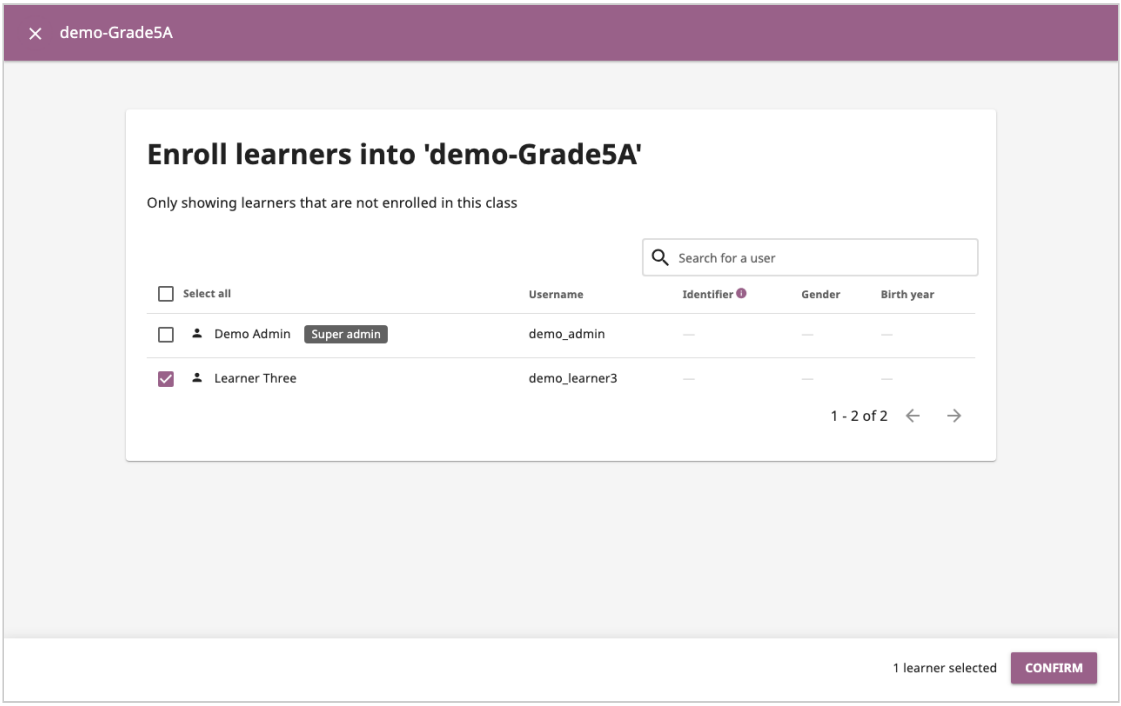


Figure 7. Learner ticked

3. The learner now appears under Learners.

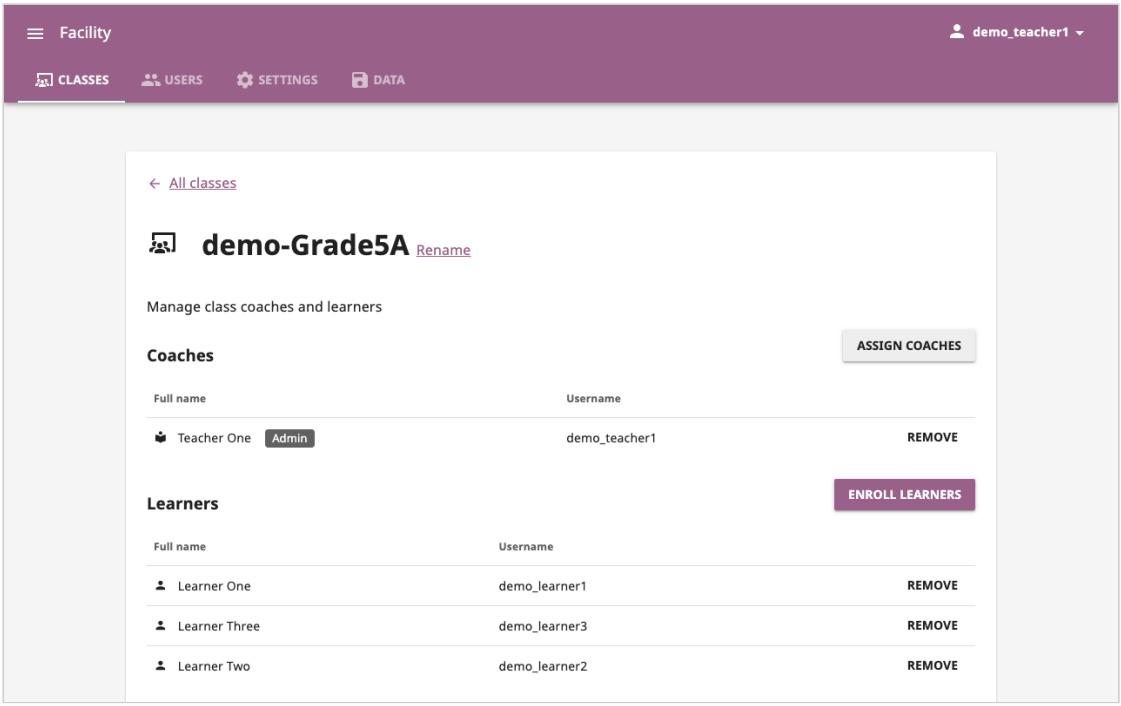


Figure 8. Learner enrolled

3.4 Make a lesson and show it to the class

A good lesson has 3 videos (explain) and 3 exercises (practise).

1. Open **Coach**, choose the class, click **PLAN > LESSONS > NEW LESSON**.

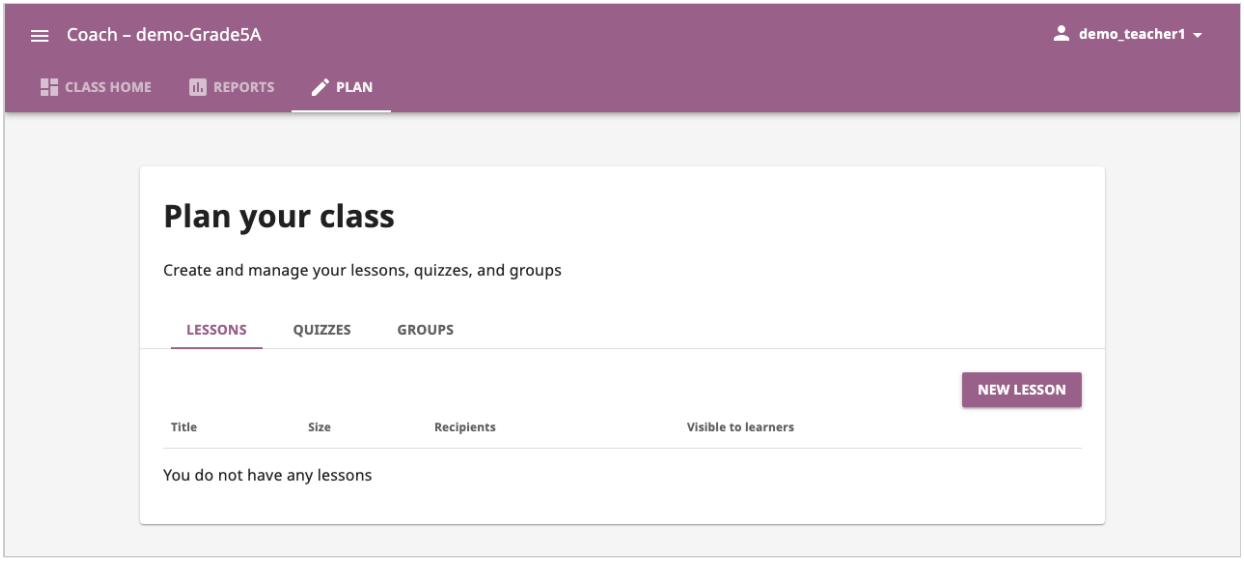


Figure 9. Plan your class

2. Type a title (`Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**.

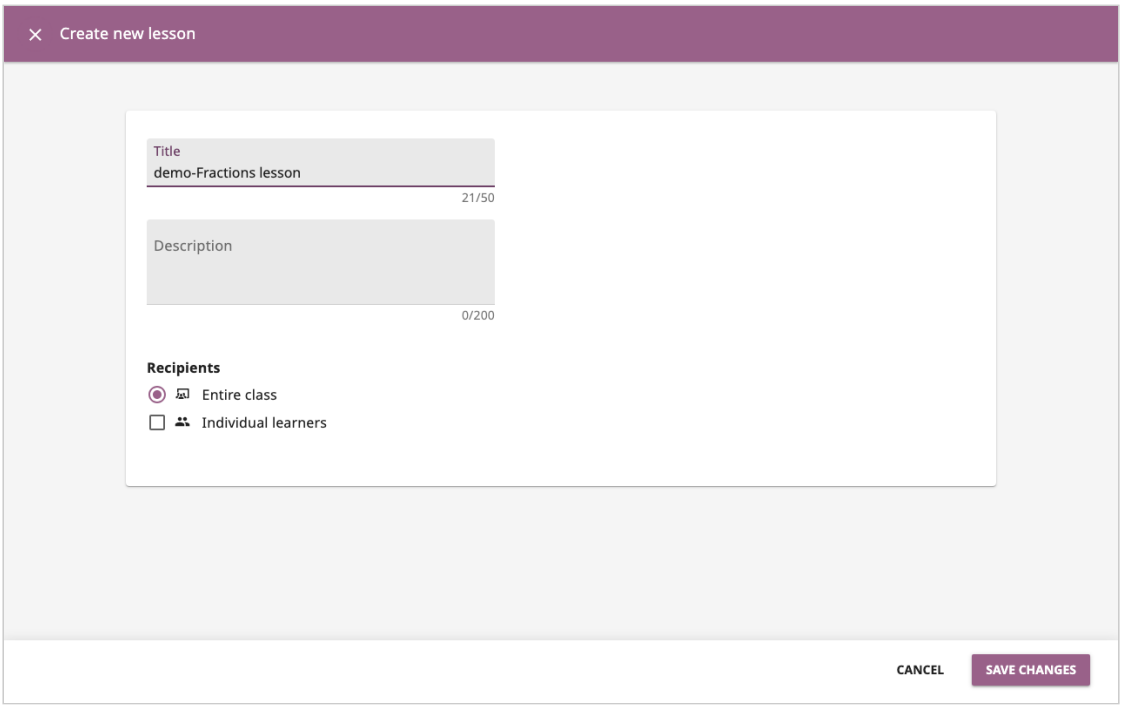


Figure 10. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

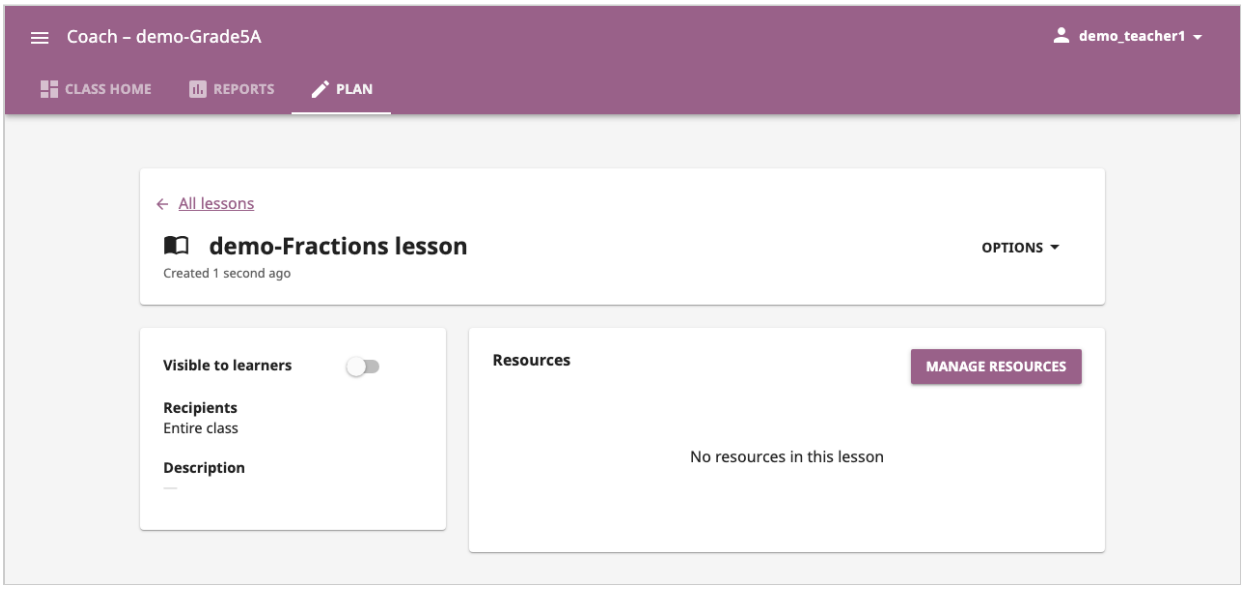


Figure 11. Empty lesson

4. Choose the channel.

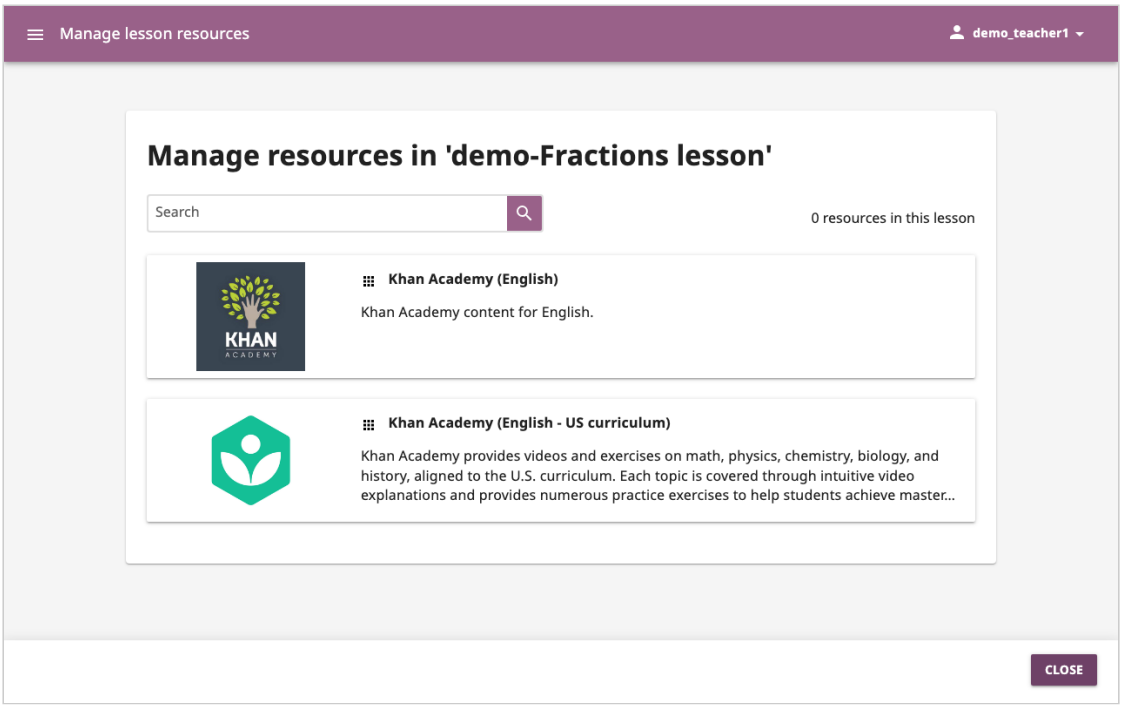


Figure 12. Channels

5. Click through **Math by grade > 5th grade > Add and subtract fractions**.

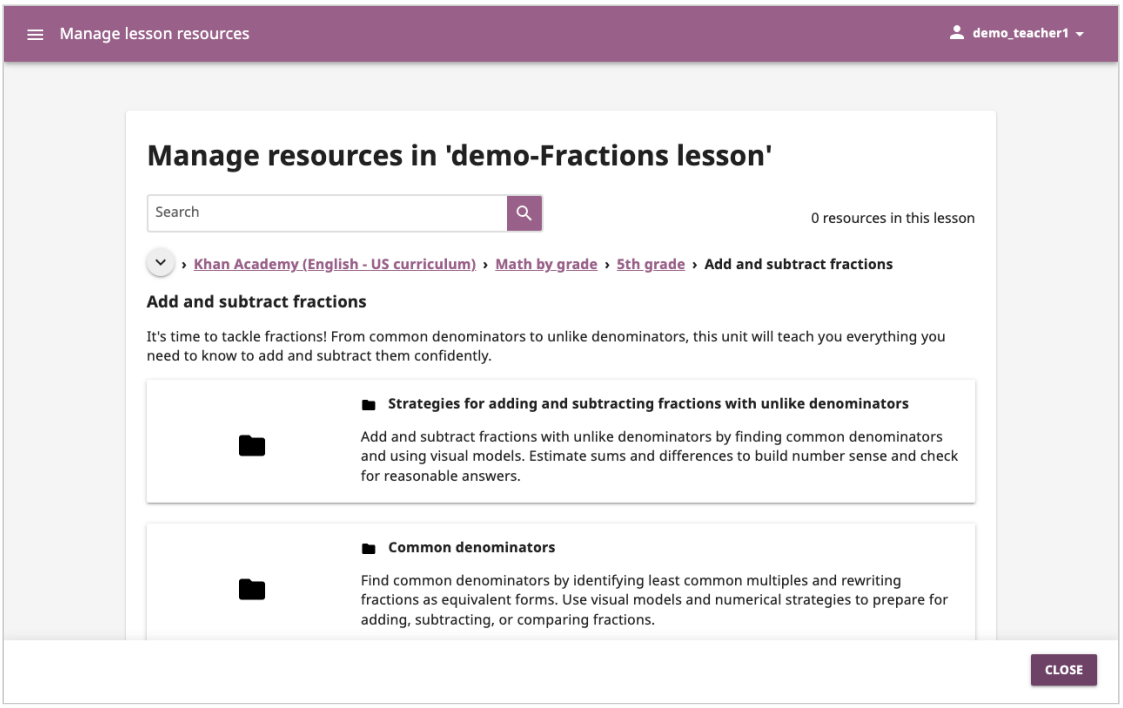


Figure 13. Topic

6. Open "Adding and subtracting fractions with unlike denominators". Tick 3 videos and 3 exercises (the exercise titles start with "Practice"). Each tick is saved straight away; a message "resources added" appears. Click **CLOSE** when done.

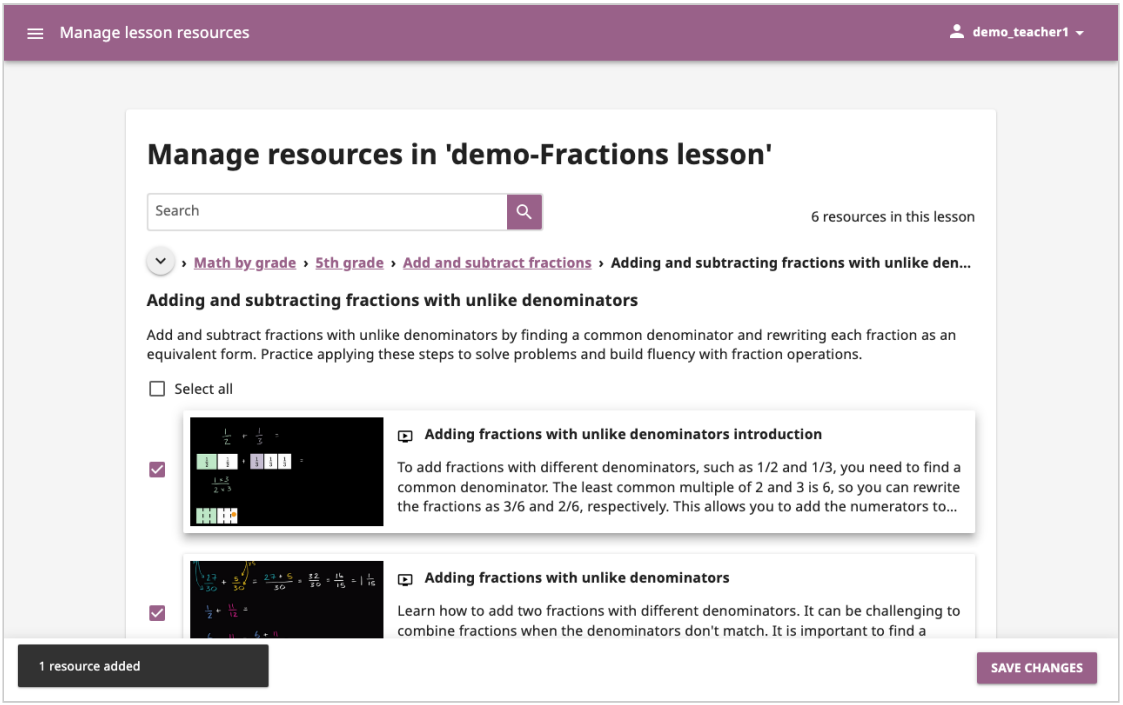


Figure 14. Resources ticked

7. Back on the lesson page, check the 6 resources and switch on **Visible to learners**.

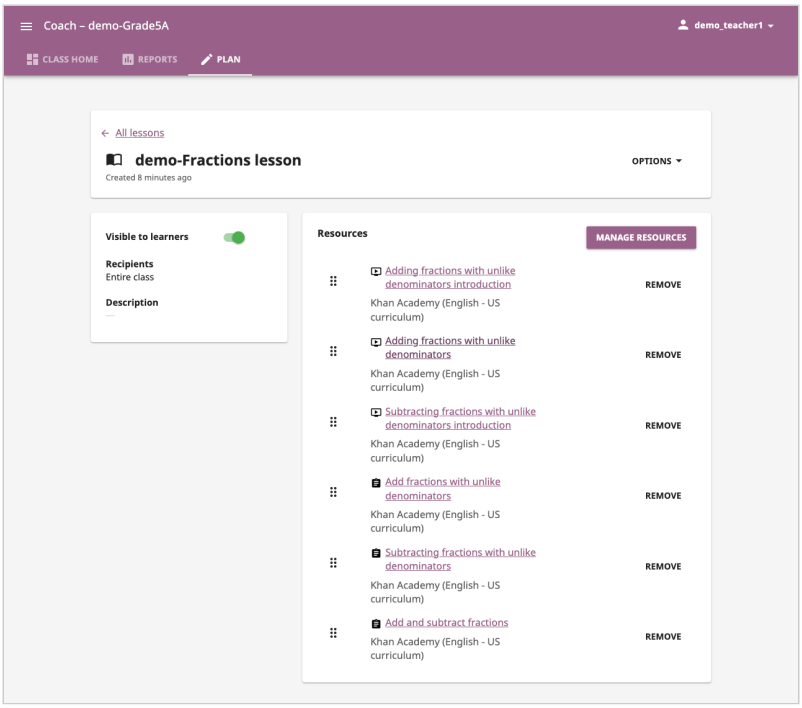


Figure 15. Lesson with 6 resources, visible

3.5 Make a quiz

Quizzes are built from exercises only.

1. Click **PLAN > QUIZZES > NEW QUIZ**.

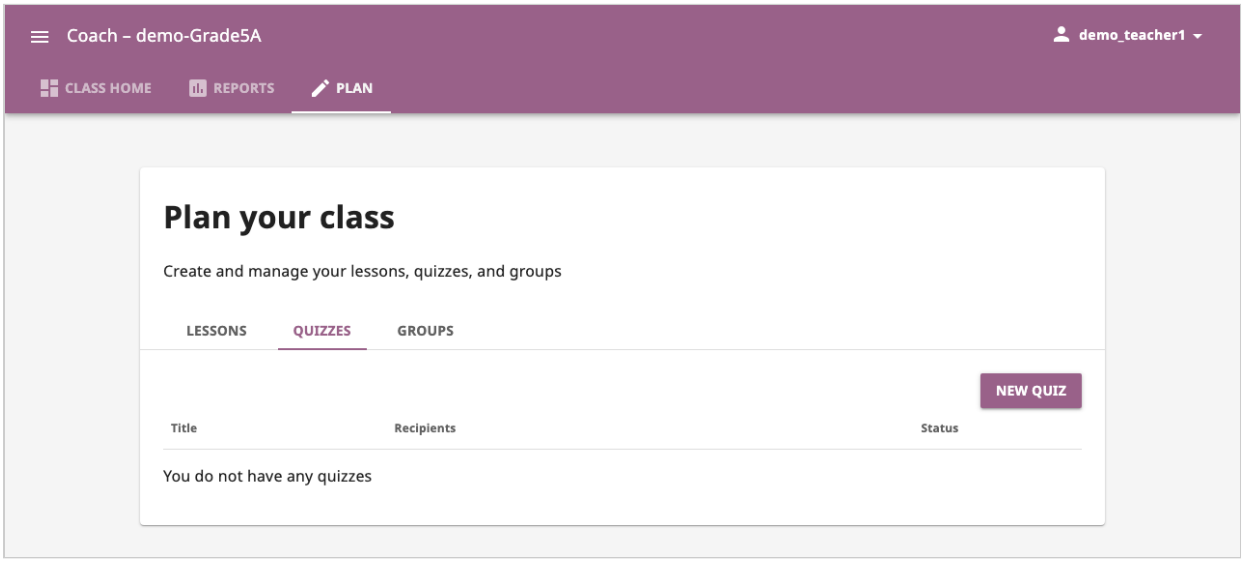


Figure 16. Quizzes tab

2. Type a title (`Fractions quiz`) and the **Number of questions** (6). Browse to Add and subtract fractions and tick a folder (or single exercises). Click **CONTINUE**.

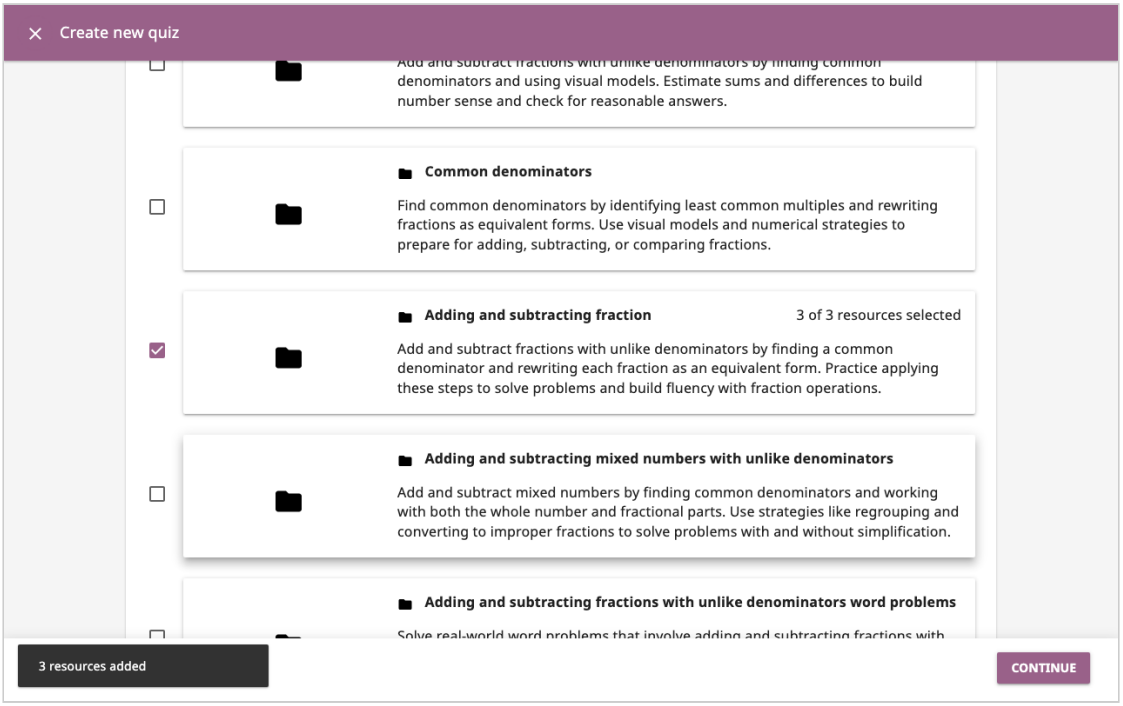


Figure 17. Create new quiz

3. **Preview quiz**: check the questions, choose **Randomized** or **Fixed** order, click **FINISH**.

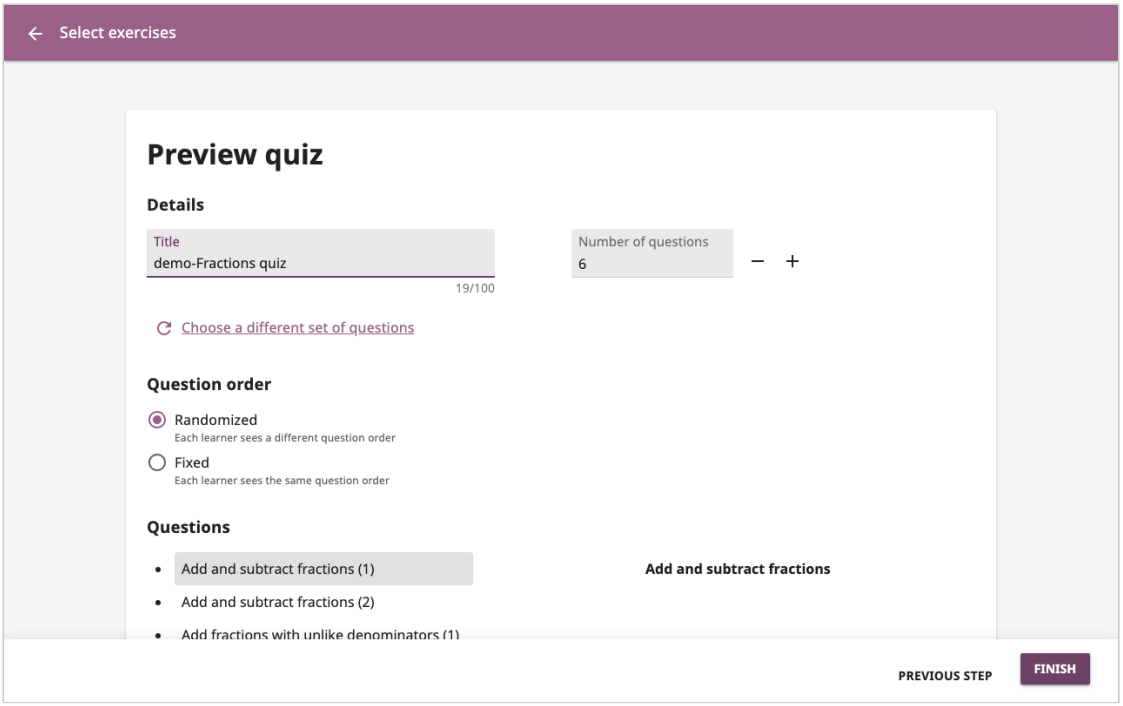


Figure 18. Preview quiz

4. The quiz is not open yet. When the class is ready, click **START QUIZ** and then **CONTINUE**. When the class has finished, open the quiz in **Coach** again and end it there, so no more answers come in.

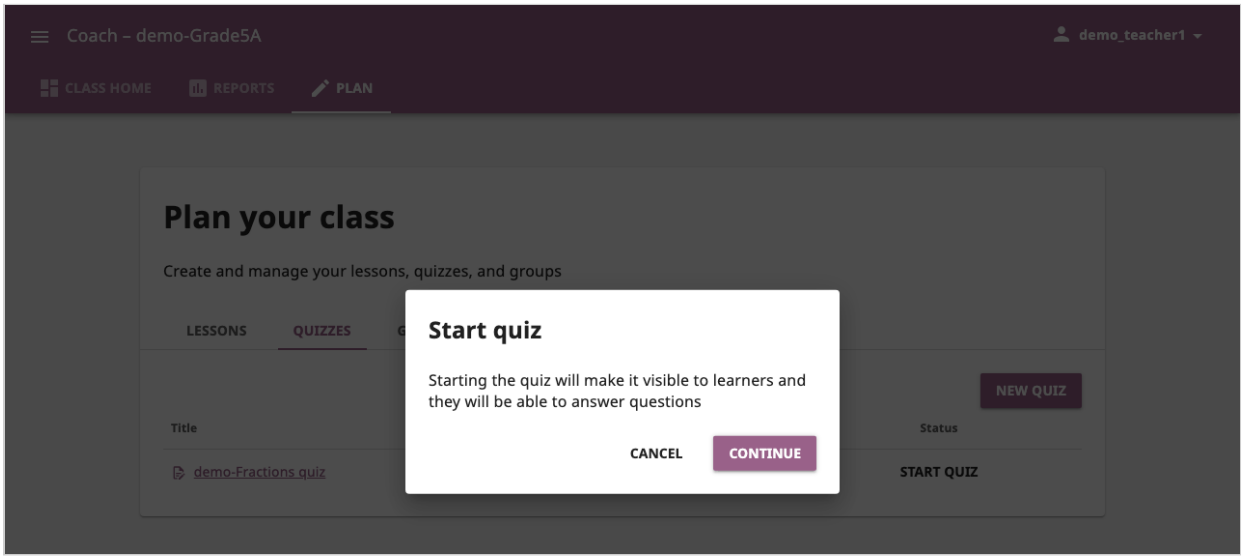


Figure 19. Start quiz

3.6 See progress (Reports)

Do this after learners have worked (chapter 4).

1. Click **REPORTS**. Tabs: **LESSONS**, **QUIZZES**, **GROUPS**, **LEARNERS**.

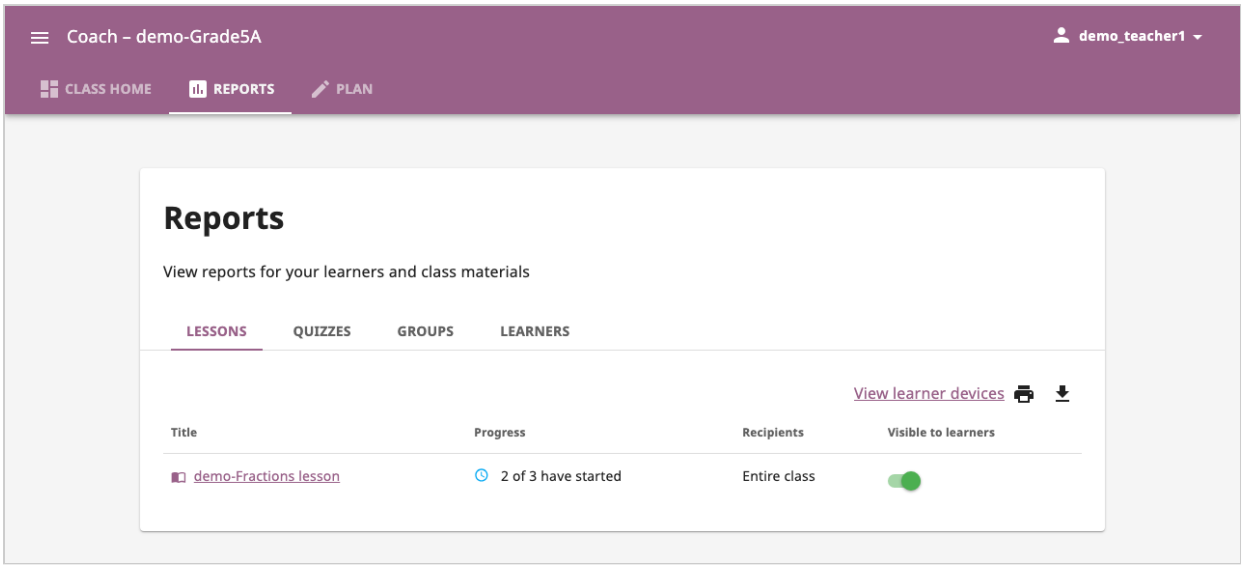


Figure 20. Reports

2. **LESSONS > (the lesson) > LEARNERS**: who completed, started or did not start.

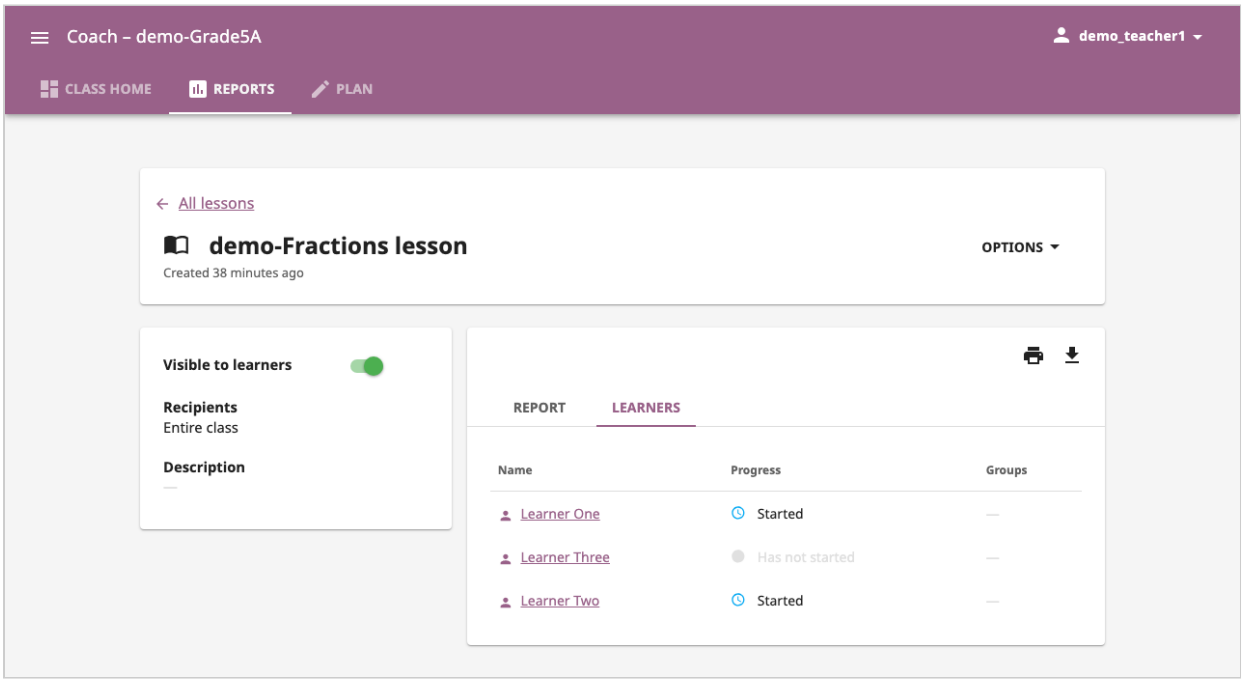


Figure 21. Lesson report

3. **QUIZZES > (the quiz)**: status and score per learner; **DIFFICULT QUESTIONS** shows what to re-teach.

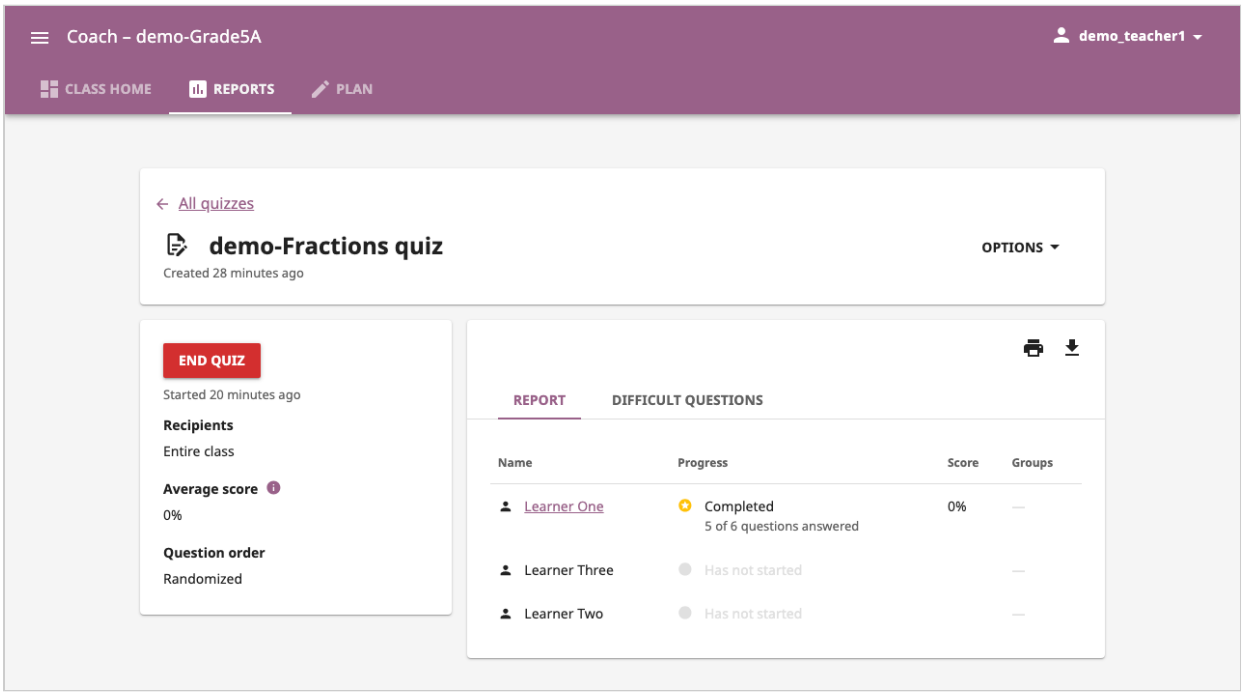


Figure 22. Quiz report

4. **LEARNERS > (a name)**: that learner's lessons and quizzes.

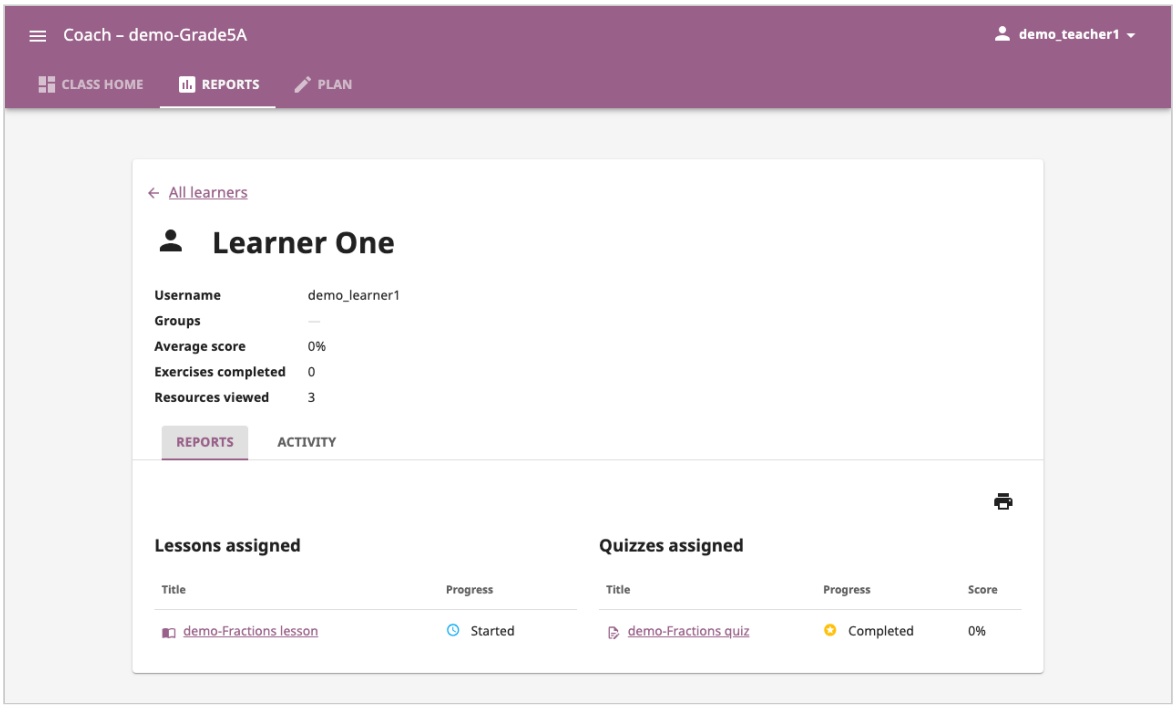


Figure 23. Learner report

Reports can be printed or exported as CSV (icons top right).

> **Tip.** Do not chase 100%. Look for the question most learners got wrong and teach it again.

4. Kolibri for learners (try it yourself)

Your learners open Kolibri the same way you do: on a classroom computer they double-click **IDEA** on the desktop and click **Open** on the Kolibri card in the Console (2.1). Show them this once.

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. In the demo school, learners sign in with a username and password: `learner1` with `learner1911!`, and so on (the IDEA default is the account name followed by `911!`; check your school's setting).

> **Note on passwords.** The demo learners have a password. A school can turn learner passwords off or on: sign in as a Kolibri admin, open **Facility > SETTINGS**, tick or untick **Require password for learners**, then click **SAVE CHANGES**. Learners who have no password yet then need one set by the admin; the IDEA default is the account name followed by `911!`.

4.1 Do a lesson

1. Sign in as `learner1`. **HOME** shows **Your classes**, **Recent lessons** and **Recent quizzes**.

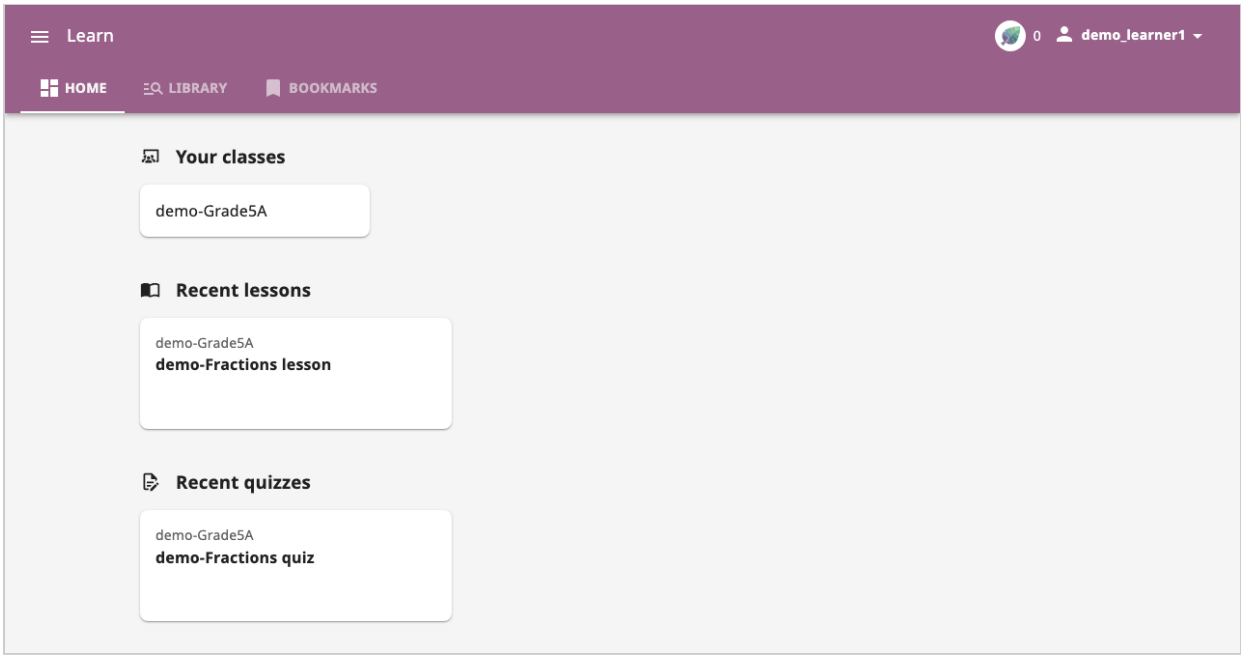


Figure 24. Learner home

2. Click **Fractions lesson**. The 6 items are listed in order.

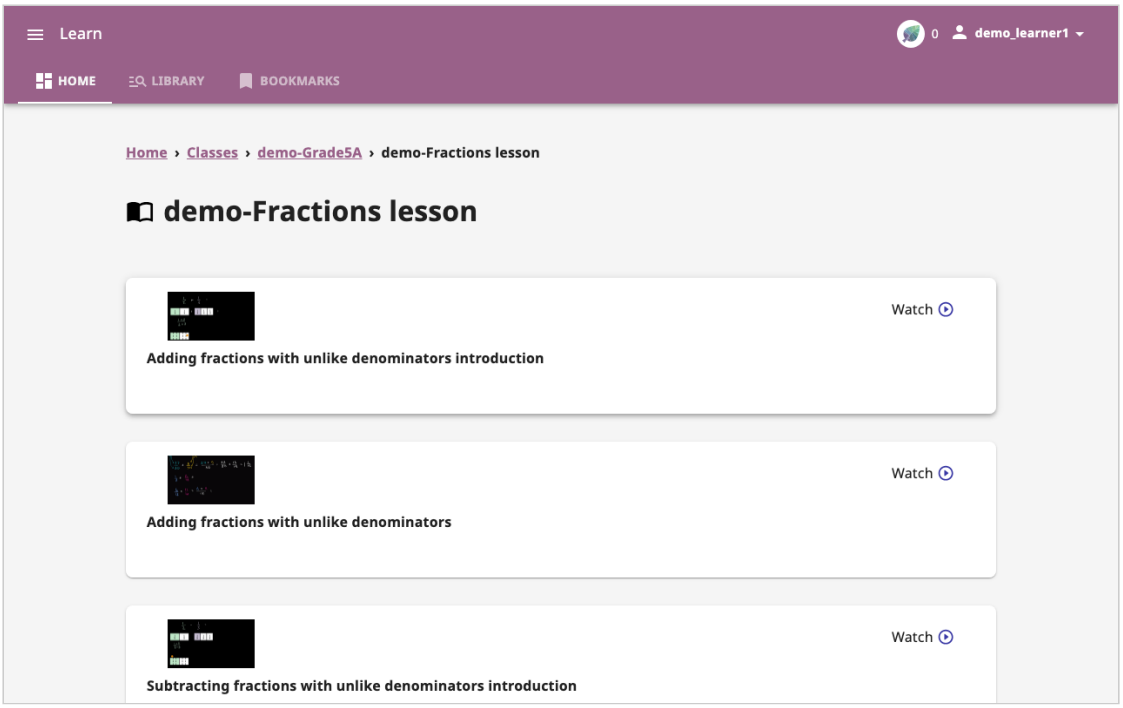


Figure 25. Lesson

3. Watch the first video.

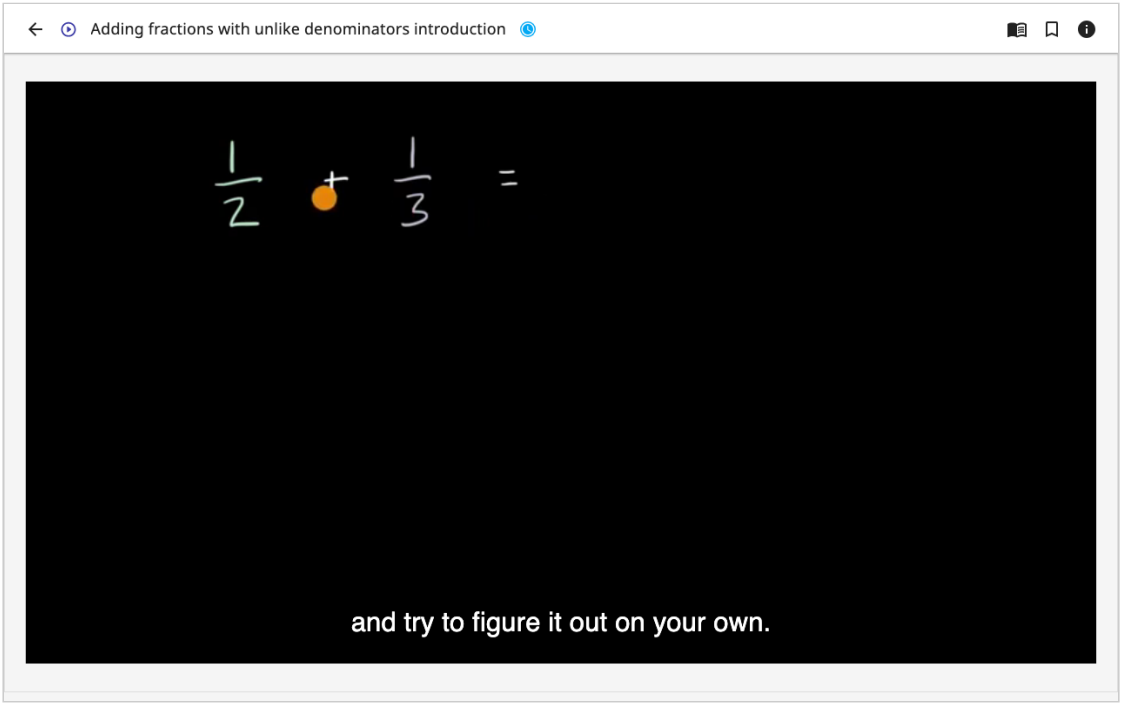


Figure 26. Video

4. Open an exercise, type an answer, click **CHECK**. Keep going until the exercise is complete ("5 correct" goal shown at the top).

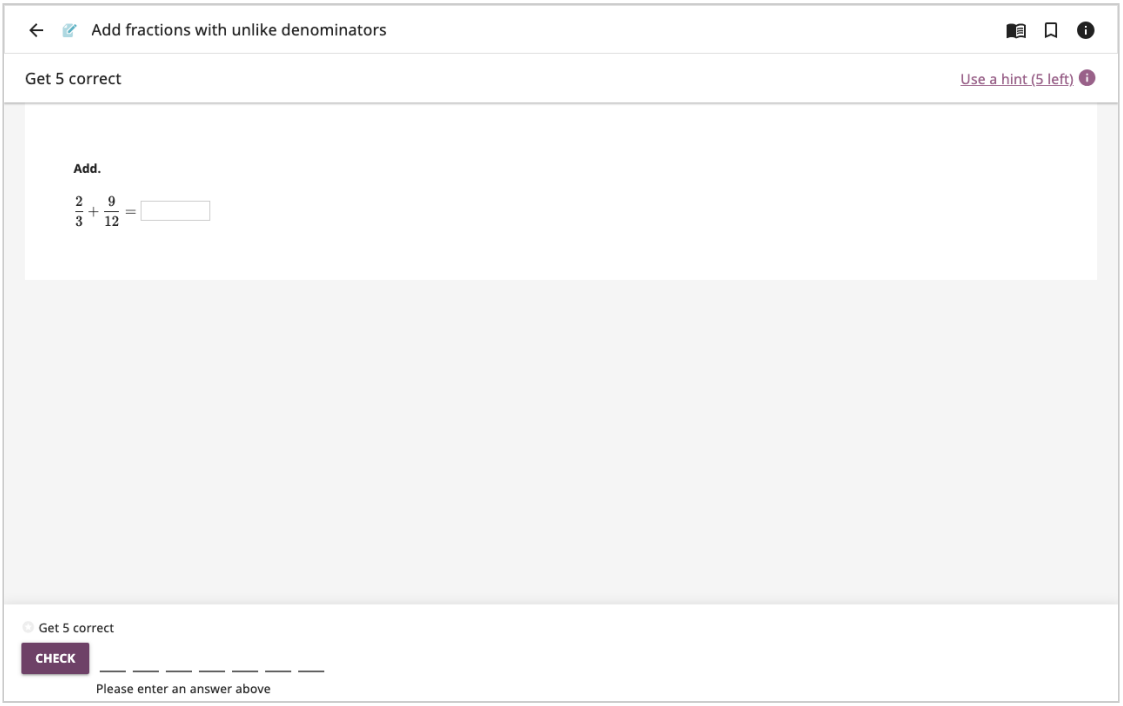


Figure 27. Exercise

5. Sign out. Sign in as `learner2` and open only one video, so the report shows a difference.

4.2 Take a quiz

1. Open the quiz from **Recent quizzes** (or the class page).

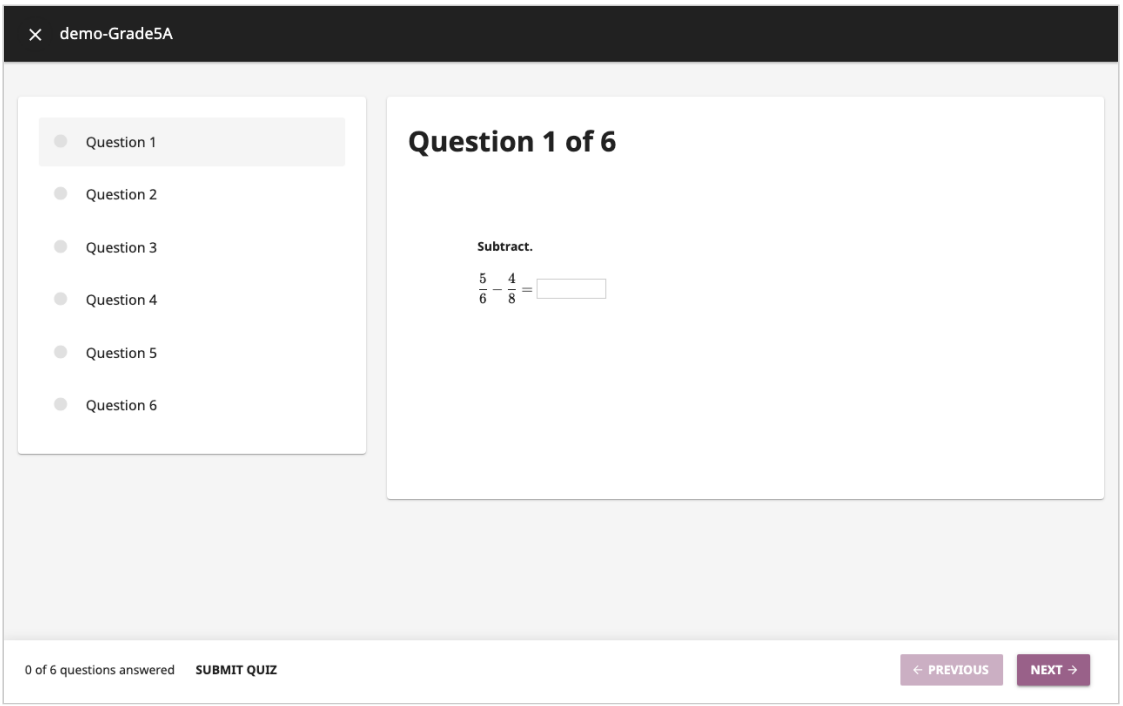


Figure 28. Quiz

2. Answer each question and use **NEXT** / **PREVIOUS**. The list on the left shows answered questions.

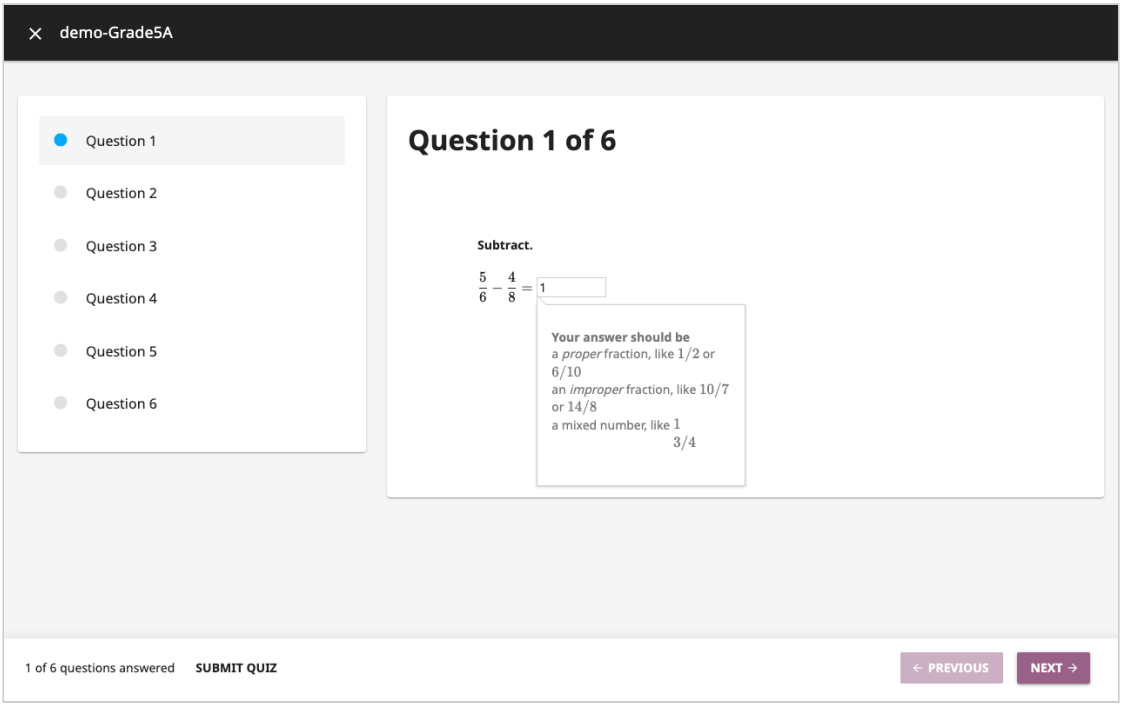


Figure 29. Question

3. Click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed.

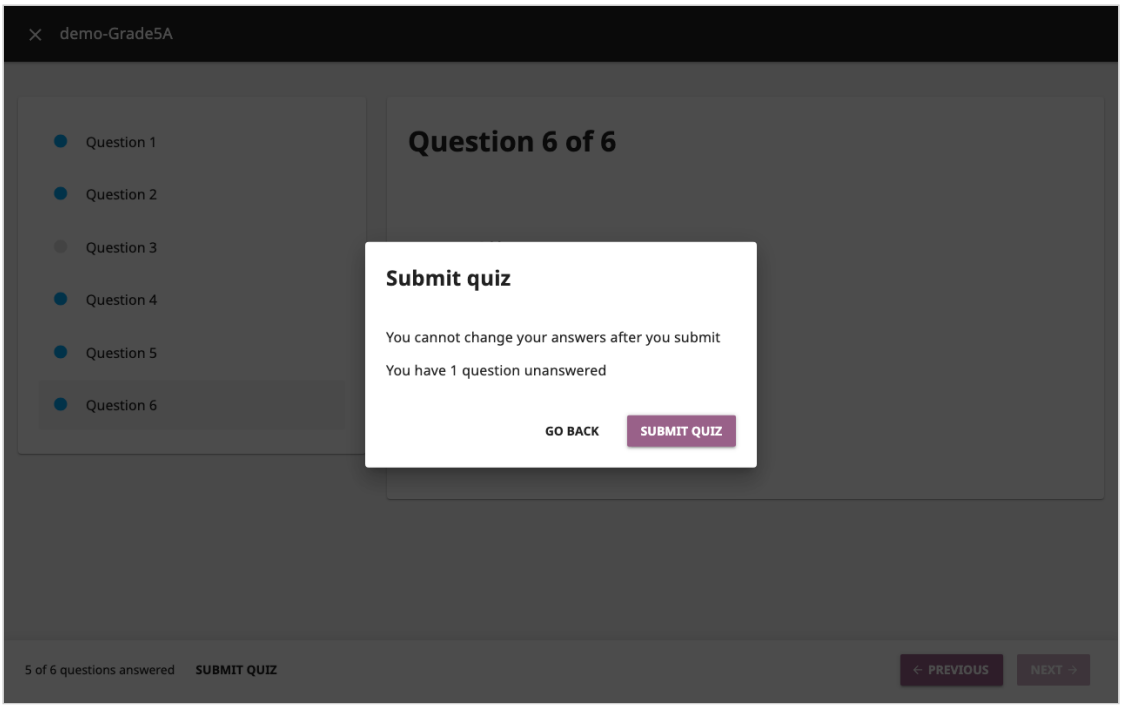


Figure 30. Submit

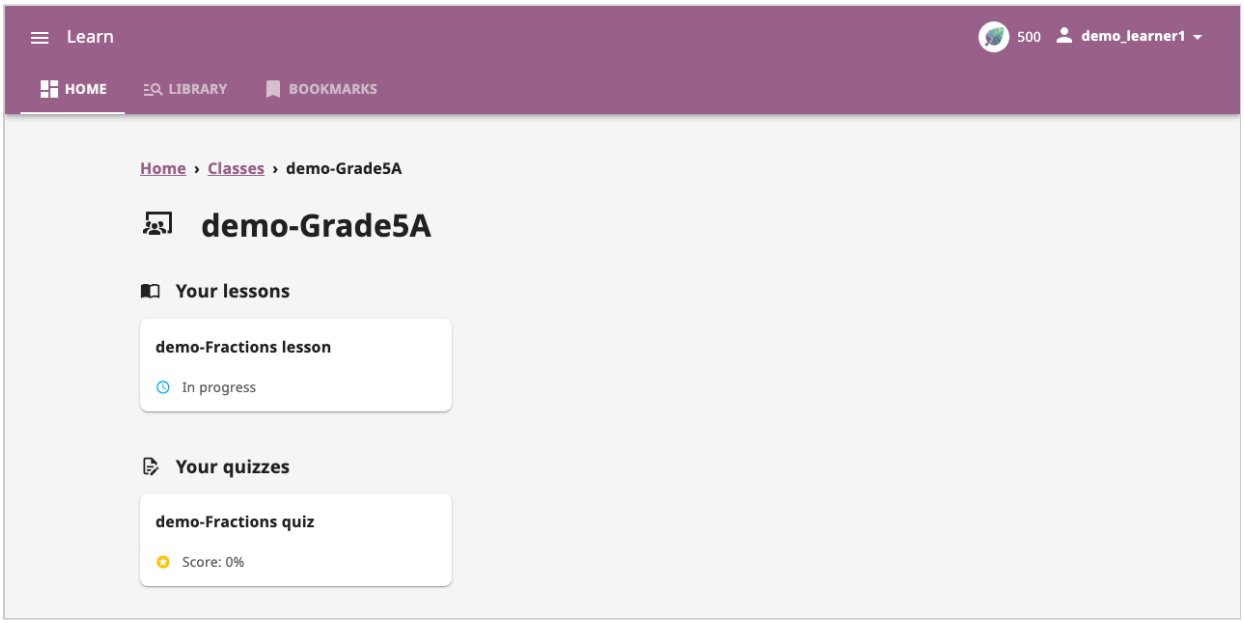


Figure 31. After submit

Now go back to the teacher window and look at Reports (3.6).

5. Make your own Kolibri channel: the channel builder and your USB stick

This is the most useful chapter for teachers. The **channel builder** turns your own videos, PDFs and pictures into a **Kolibri channel**. Your learners then find it in Kolibri next to the Khan Academy lessons, and you can put it in lessons and quizzes (chapter 3).

You do not need internet or an account. Bring your files on a normal **USB stick**, filled at home, in town or from a colleague.

Good to know:

- **IDEA only reads your stick.** It never writes to it, so nothing on the stick changes, and the stick still works in any Windows laptop afterwards. You do not have to prepare or erase it.
- The stick must be **FAT32 or exFAT**. Almost all sticks are, unless someone formatted them as NTFS.
- **Building is not publishing.** A channel reaches learners only after someone imports it in Kolibri (5.2).

5.1 From your USB stick to a channel (automatic)

At home or in town (any computer):

1. Open the stick and make a folder named **`kolibri`** at the top level of the stick.
2. Inside `kolibri`, make **one folder per channel**, for example `Grade 5 Science`. The folder name becomes the channel name.
3. Inside the channel folder, make **one folder per topic**, for example `Plants` and `Water cycle`, and copy your videos and PDFs into them.

> **Screenshot pending retake (C5-01):** The stick in Windows File Explorer: kolibri > Grade 5 Science > Plants, Water cycle.

> **Optional: `channel.json`.** To set the channel's name, author, description or licence yourself, put a small text file `channel.json` into the channel folder (your IDEA contact can give you an example). Without it, the channel gets the folder name, the stick's name as author, and the standard licence.

At school:

4. Plug the stick into a free USB port of the IDEA box.
5. The IDEA box starts the channel builder by itself, if the channel builder is on this box. It then builds **one channel for each folder inside `kolibri`**, one after the other. You do not have to click anything.
6. To watch it, open the Console. Under the IDEA box you see the row **USB stick: (your stick's name) (read-only)**. Click the link to the channel builder, or open **Channel builder** in the app list.

Screenshot pending retake (C5-02): Console: the USB stick row under the IDEA box, with the link to the channel builder.

7. On the builder's start page the box **From USB sticks** shows each channel: **Waiting**, then **Building** with the number of files done, then **Built**.

Screenshot pending retake (C5-03): Channel builder, From USB sticks box: Building (files done out of total).

8. When it says **Built, you can pull the stick out**, you can take your stick out. The channel stays on the IDEA box.

Screenshot pending retake (C5-04): Channel builder, From USB sticks box: Built, you can pull the stick out.

Don't pull the stick out while it says Building. If you do, nothing breaks: the build stops with the message "stick removed before the build finished; plug it in again". Plug the stick in again and the build starts over.

5.2 Import the channel into Kolibri

Your channel is built, but learners cannot see it yet. A Kolibri admin imports it once. On the demo school, `demo_teacher1` is a Kolibri admin.

1. Open Kolibri from the Console and sign in as a Kolibri admin.
2. Open **Device**, then **Channels**, and click **IMPORT**.

Screenshot pending retake (C5-05): Kolibri Device > Channels with the IMPORT button.

3. Choose **Local network** (not Internet, not a local drive).

Screenshot pending retake (C5-06): Kolibri import: choose Local network.

4. Pick the **channel builder** from the list of devices on the network.

Screenshot pending retake (C5-07): Kolibri import: the channel builder in the list of network devices.

5. Pick your channel (for example **Grade 5 Science**) and click **Import**.

Screenshot pending retake (C5-08): Kolibri import: the built channel listed, Import button.

6. Choose the topics (or everything) and click **IMPORT**. Wait until the task says it is finished.

Screenshot pending retake (C5-09): Kolibri import finished; the channel in the Channels list.

7. Now use it like any other channel: add its videos and PDFs to a lesson (3.4) or a quiz (3.5). Learners also find it under **Explore** (or **Library**).

Screenshot pending retake (C5-10): Learner view: the new channel next to the other channels.

5.3 Update the channel

1. Next term, add or change files **in the same folder** on your stick (for example a new topic folder `Soil`).
2. Plug the stick in again. The builder sees what changed and builds a new version of the **same channel**. If nothing changed, it does nothing.
3. In Kolibri, open **Device > Channels**. Your channel shows that an update is available. Import the update the same way as in 5.2. Lessons that use the channel keep working.

Screenshot pending retake (C5-11): Kolibri Device > Channels: update available for the channel.

5.4 Build a channel by hand

Use this when your stick has **no `kolibri` folder**, or when your files are in **School Files** (chapter 9) or in your Nextcloud folders.

1. Open **Channel builder** from the Console.
2. Pick the folder to build from. Folders on a stick start with **USB:**, folders on the Files Disk with **Files Disk:**.

Screenshot pending retake (C5-12): Channel builder: folder picker with a USB: source.

3. Type the channel name and the author, and click **Build**.
4. Wait until the page says **Build finished**. Then import the channel in Kolibri (5.2).

Screenshot pending retake (C5-13): Channel builder: Build finished.

Building from the same folder again updates the same channel (5.3).

5.5 Use your stick in Nextcloud

Your stick also appears in Nextcloud, so you can open its files or copy them into your own folders, without building a channel.

1. Plug the stick in. Open Nextcloud and sign in as a teacher. In **Files** you see a folder **USB: (your stick's name)** with a lock icon: it is read-only.

Screenshot pending retake (C5-14): Nextcloud Files: the USB: folder with the lock icon.

2. Open the folder to view or play a file. To keep a file, select it, choose **Copy**, and copy it into one of your own folders (for example a course folder, or **School Files**).

Screenshot pending retake (C5-15): Nextcloud: the Copy dialog, copying from the USB folder into a course folder.

3. Pull the stick out when you are done. The **USB:** folder disappears; the files you copied stay in Nextcloud.

Only teachers see the **USB:** folder. Learners do not.

5.6 Questions

Question	Answer
My stick doesn't show up.	Is it plugged into **this** IDEA box? A stick only shows on the box it is plugged into. Is it NTFS? NTFS sticks are not read; copy the files onto a FAT32 or exFAT stick, or ask the operator.
Nothing was built.	Check that the stick has a folder named `kolibri` at the top, with one folder per channel inside it. Check that the channel builder is on this IDEA box and was not stopped by the operator (the Console row says so). A stick with more than 8 GB of files is not built automatically: build it by hand (5.4).
I saved a file, but it's not on my stick.	IDEA never writes to your stick. Download the file from Nextcloud onto a laptop and copy it to the stick there.
My learners can't see my channel.	Building is not publishing. Import the channel in Kolibri first (5.2), then add it to a lesson.
I changed my files, but the channel is the same.	Plug the stick in again, wait for **Built**, then import the update in Kolibri (5.3).

6. Nextcloud for teachers

Demo: `demo-teacher1`, groups `demo-Grade5A` (class) and `demo-teachers`. Office documents open in **Nextcloud Office**.

Open Nextcloud from the Console (chapter 2) and sign in. Click **Files** at the top.

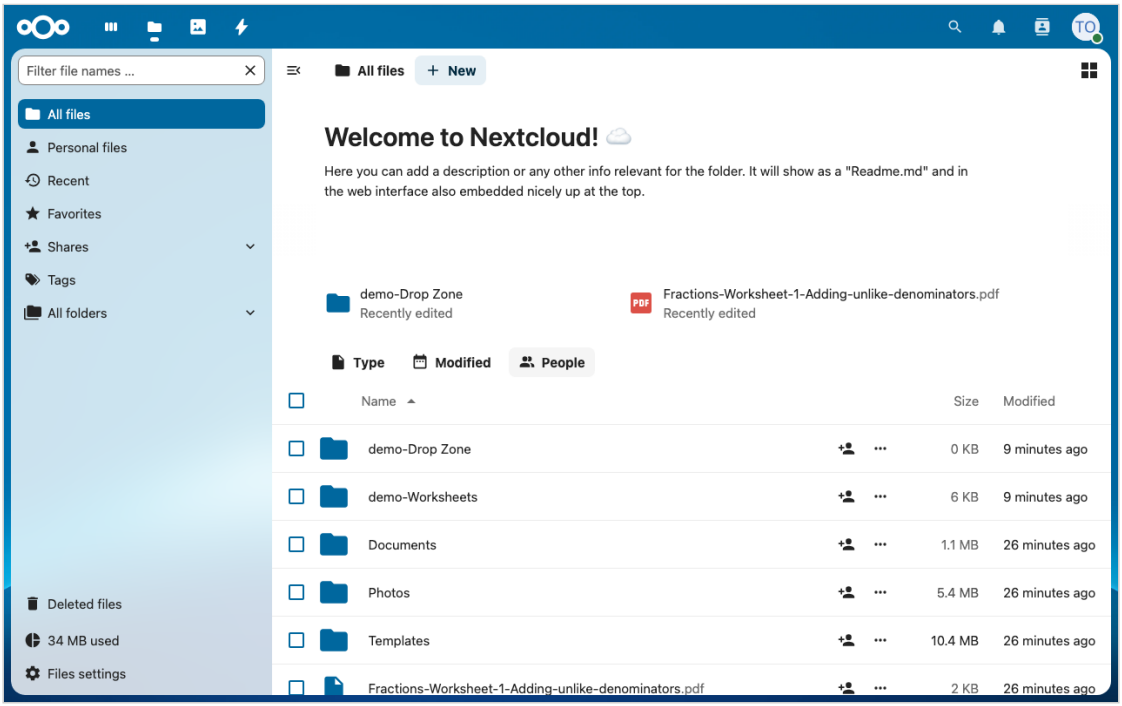


Figure 32. Files

6.1 Share worksheets with the class (view only)

1. Next to the folder `demo-Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

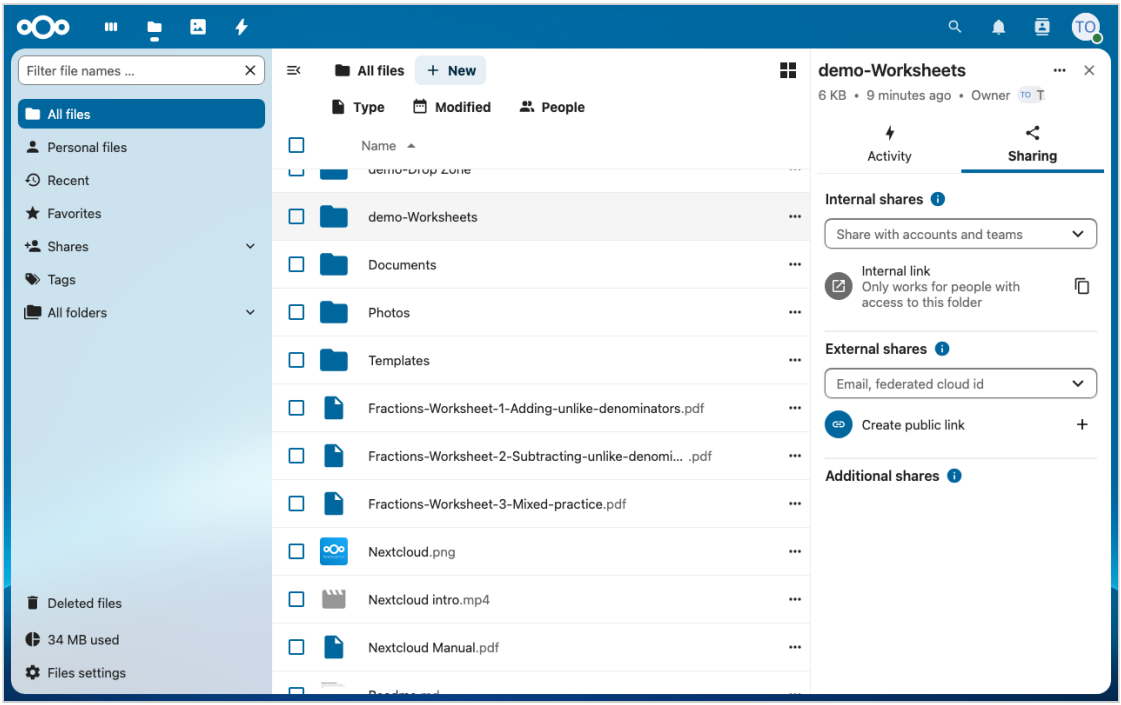


Figure 33. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type `demo-Grade5A` and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

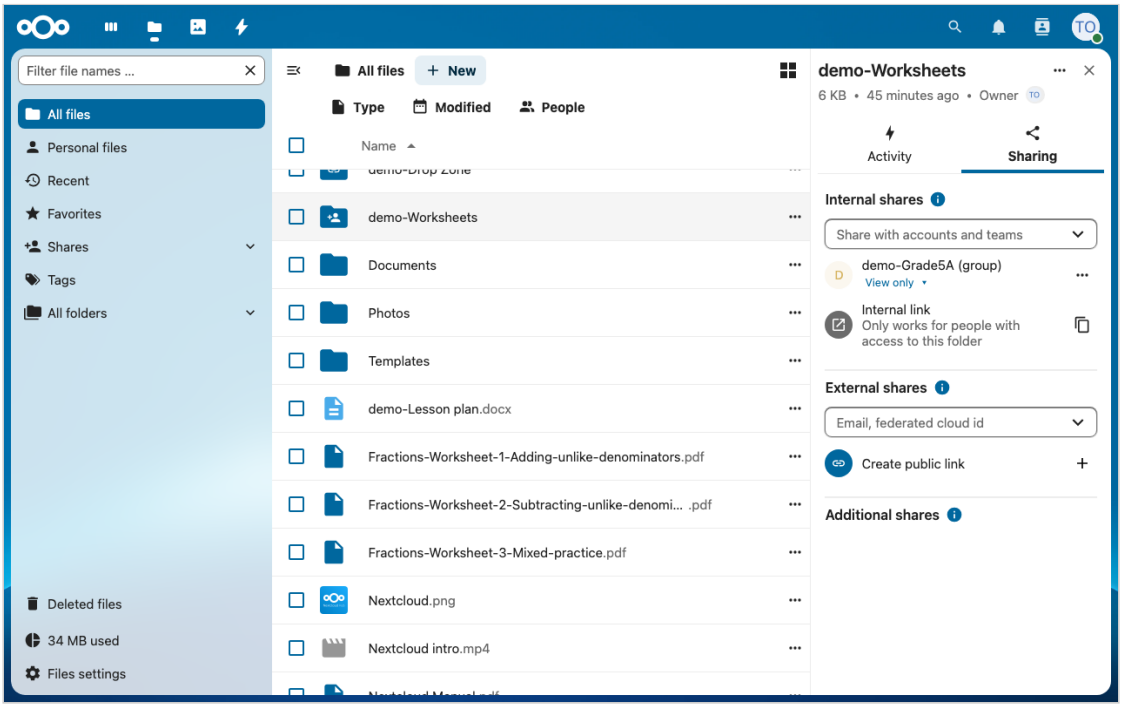


Figure 34. Shared with group

Learners now see the folder but cannot change or delete files.

6.2 Write a document together

1. Click **+ New > New document**.

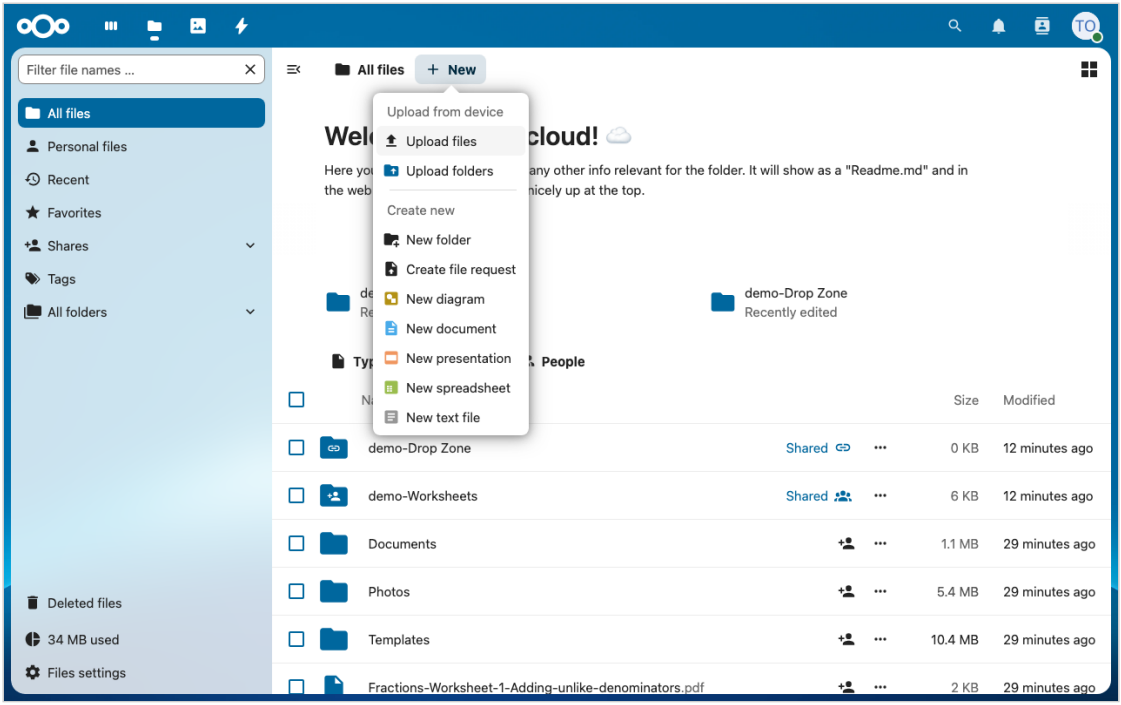


Figure 35. New menu

2. Type a name (`demo-Lesson plan`) and click **Create**.

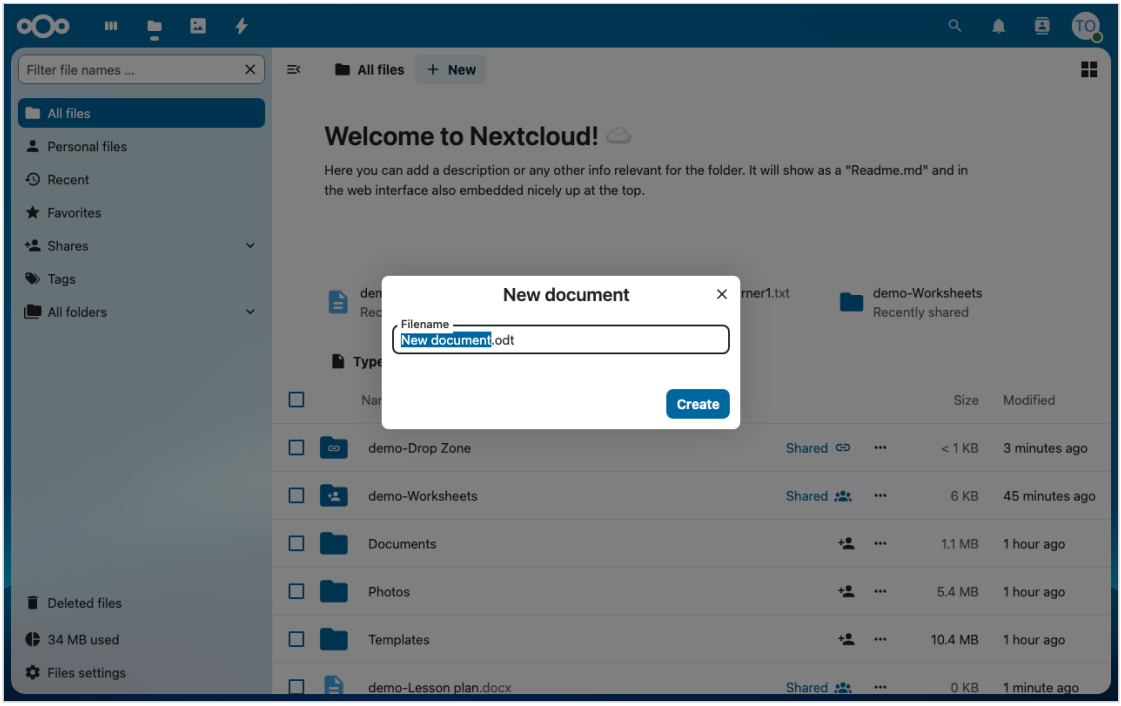


Figure 36. Name

3. Pick a template (Blank is fine) and click **Create**.

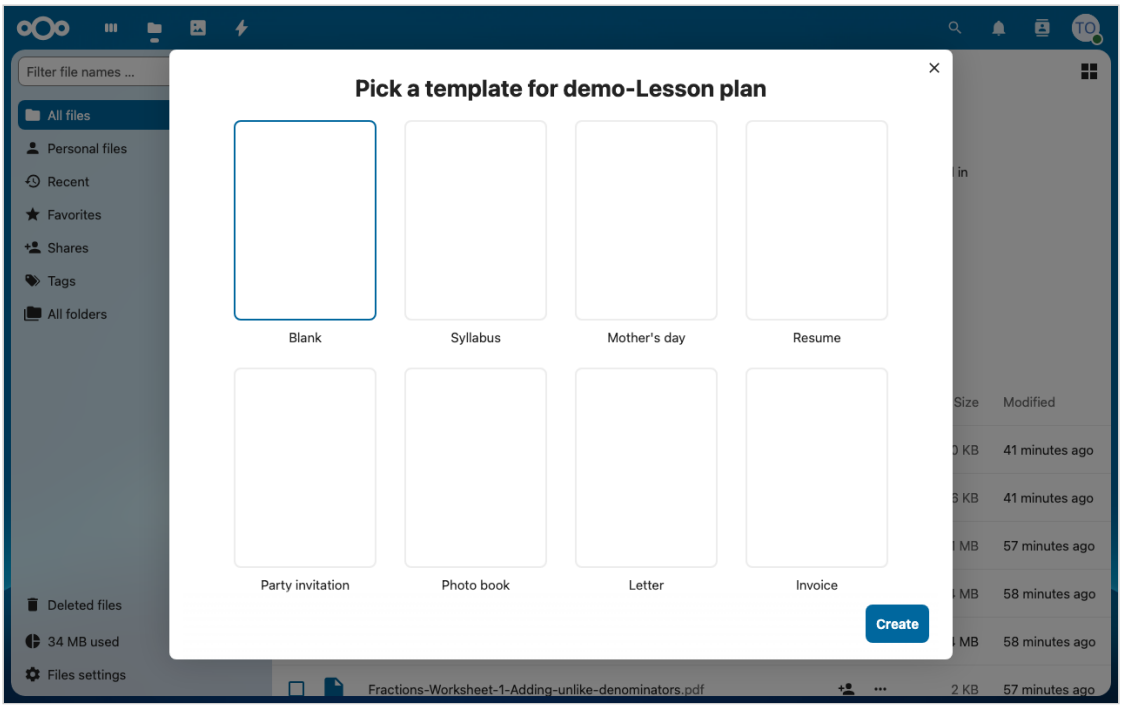


Figure 37. Template

4. Share the file with `demo-teachers` and/or your class group, with editing allowed.

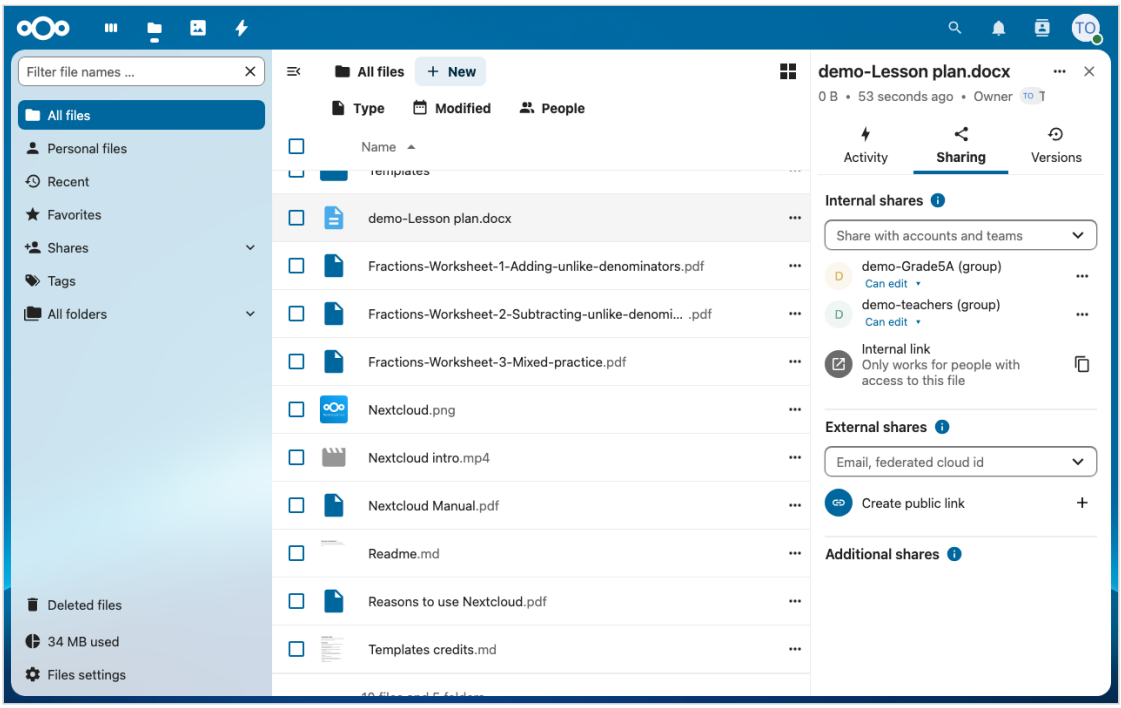


Figure 38. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

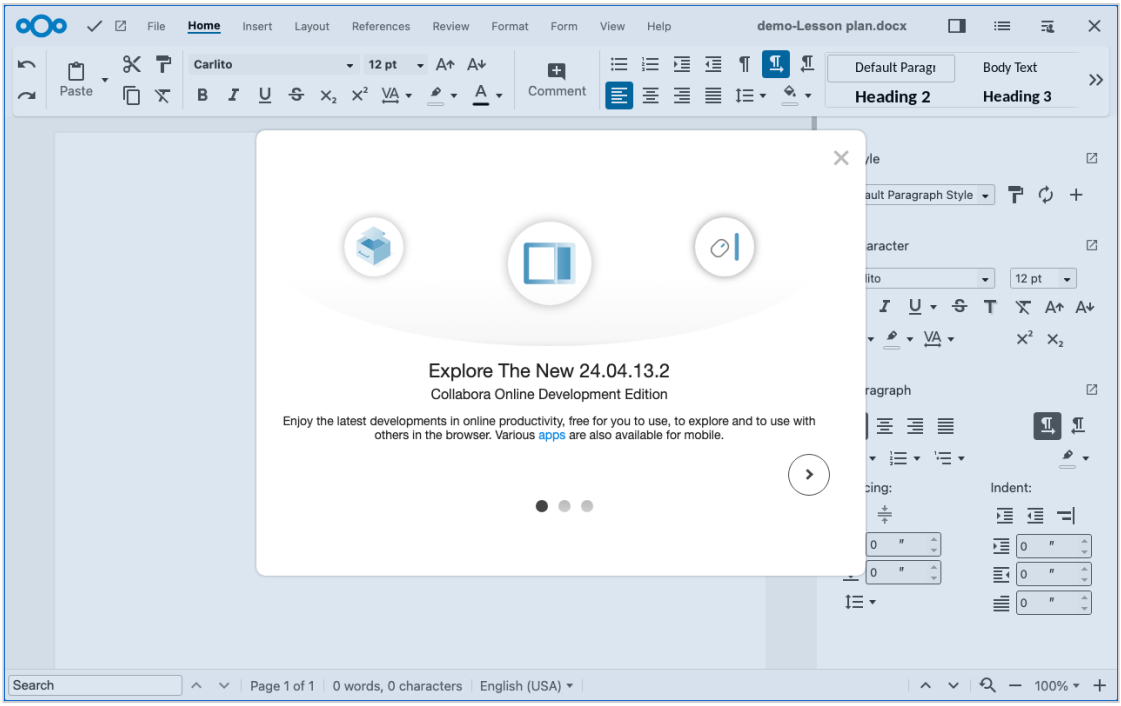


Figure 39. Nextcloud Office

6.3 Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Open sharing for folder `demo-Drop Zone`.

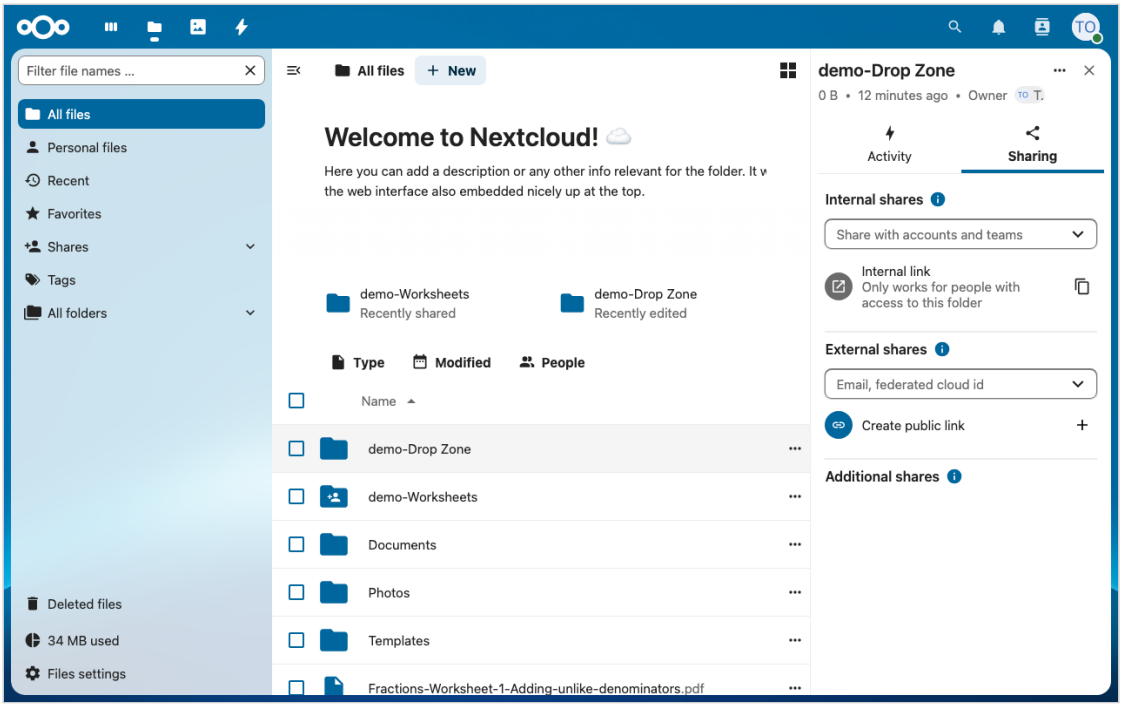


Figure 40. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

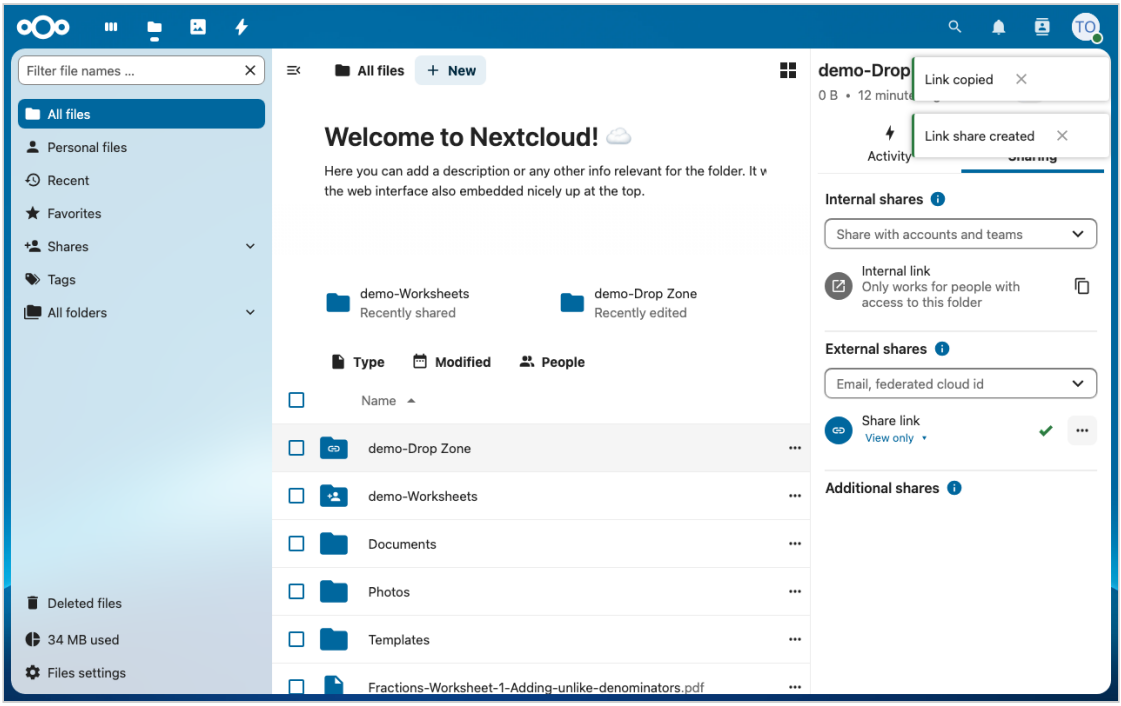


Figure 41. Create link

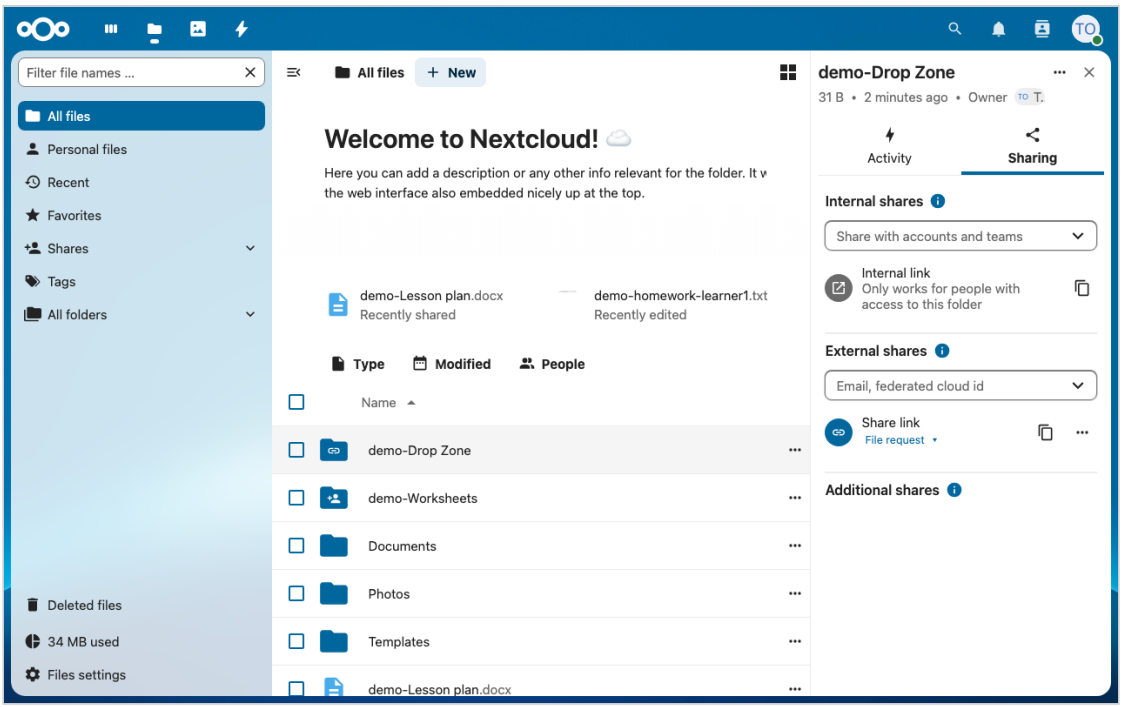


Figure 42. Link created

3. Give learners the link (for example, write it on the board). Uploaded files appear in your `demo-Drop Zone` folder.

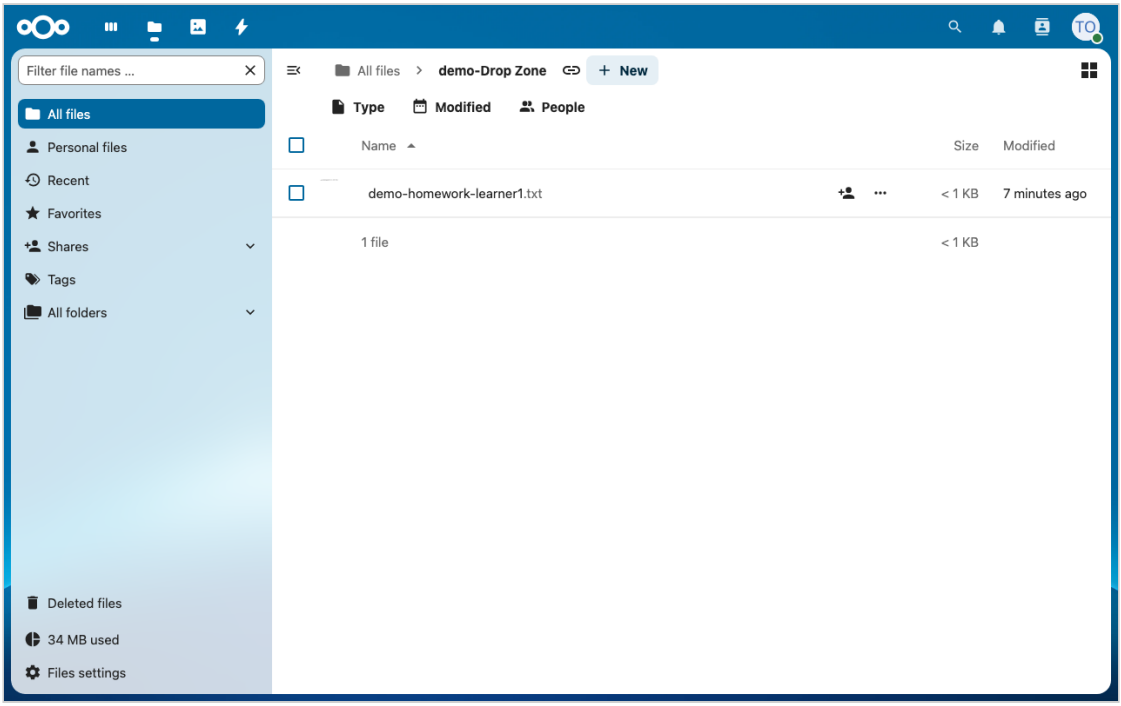


Figure 43. Received homework

7. Nextcloud for learners (try it yourself)

Learners open Nextcloud from the Console, like Kolibri: **IDEA** on the desktop, then **Open** on the Nextcloud card (2.1). They have no personal Nextcloud app on the classroom computer, so they always work in the browser.

7.1 Open class files and co-edit

1. Sign in to Nextcloud as `learner1` (private window). Shared folders and files appear in **Files** with the teacher's name.

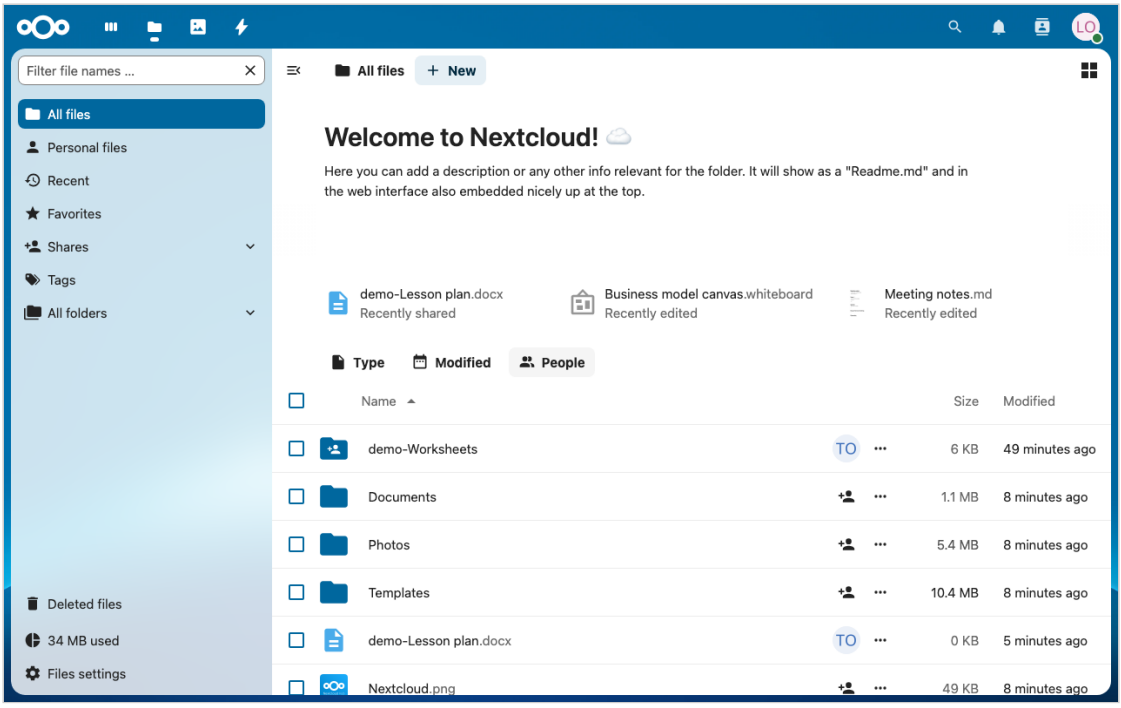


Figure 44. Learner files

2. Open `Worksheets`: the learner can open and download the worksheets, not change them.

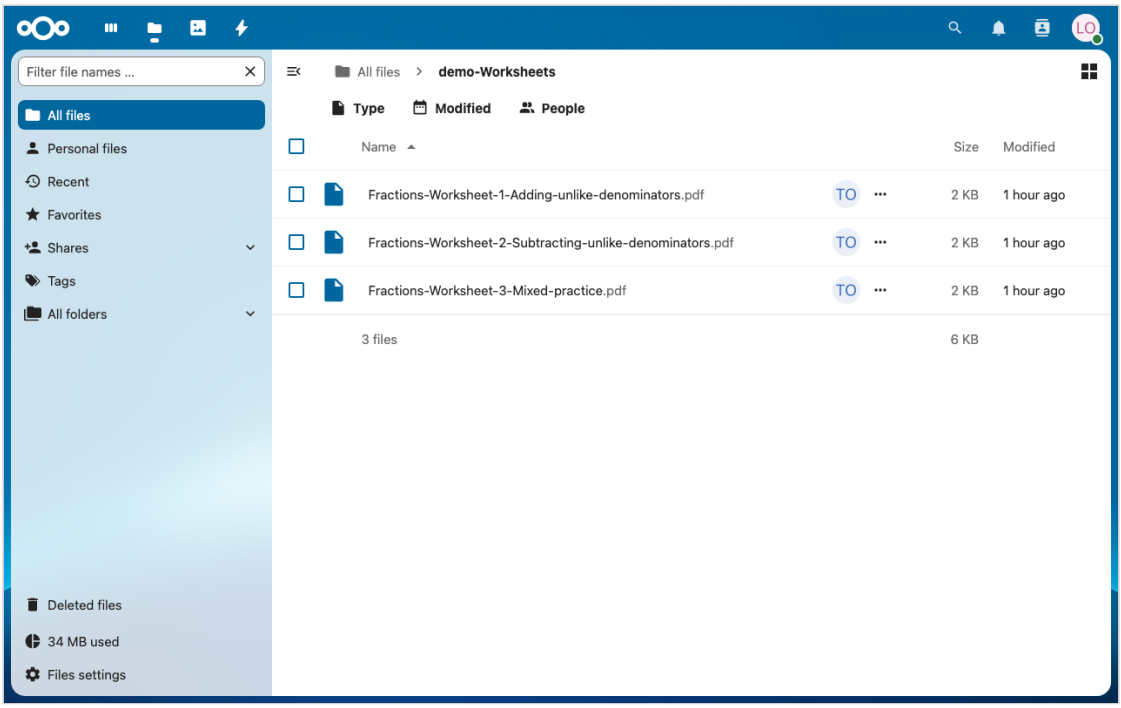


Figure 45. Worksheets

3. Open `Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

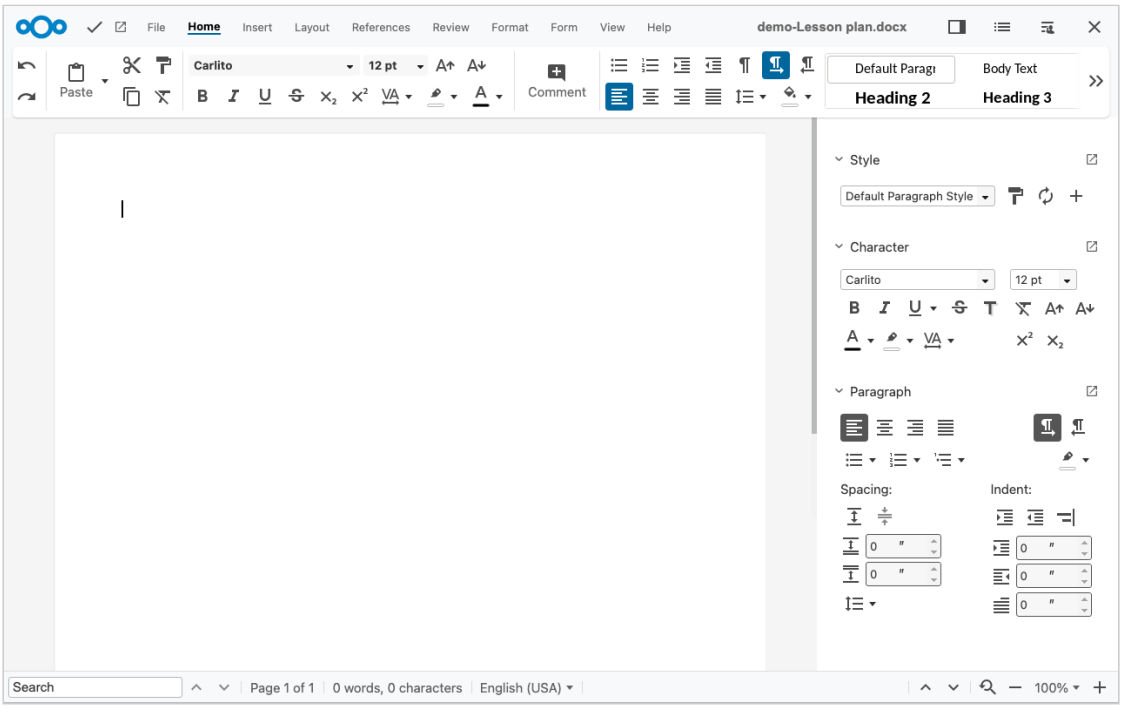


Figure 46. Learner in document

7.2 Hand in homework

1. Open the File drop link from 6.3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

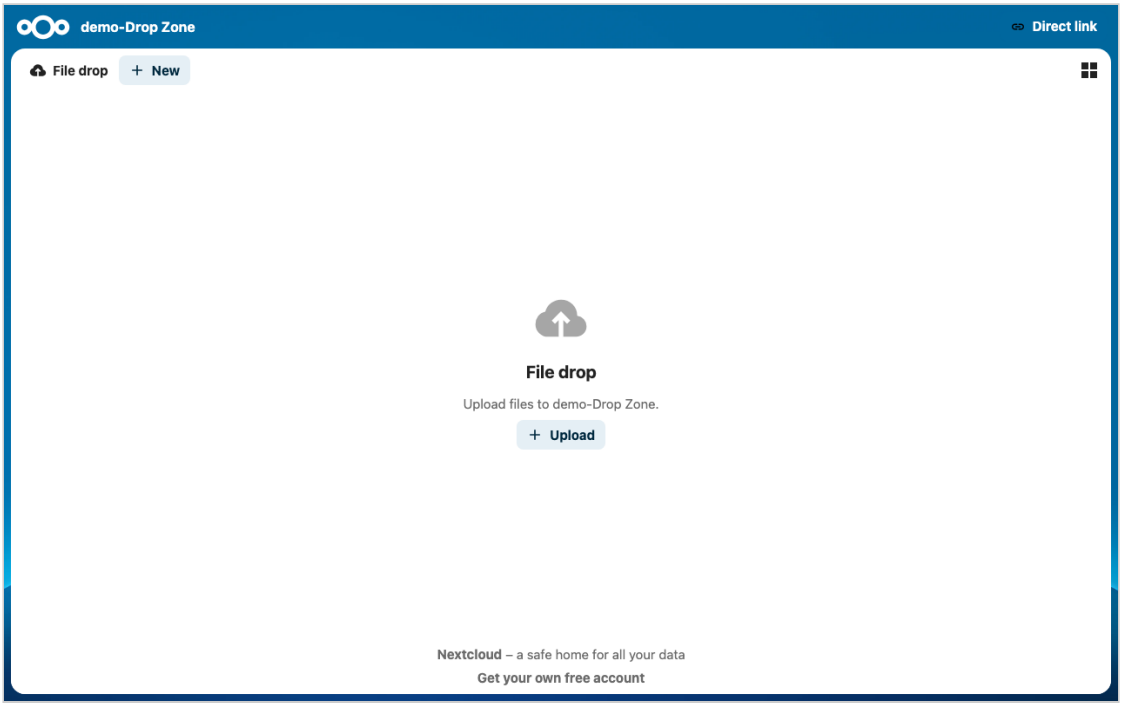


Figure 47. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

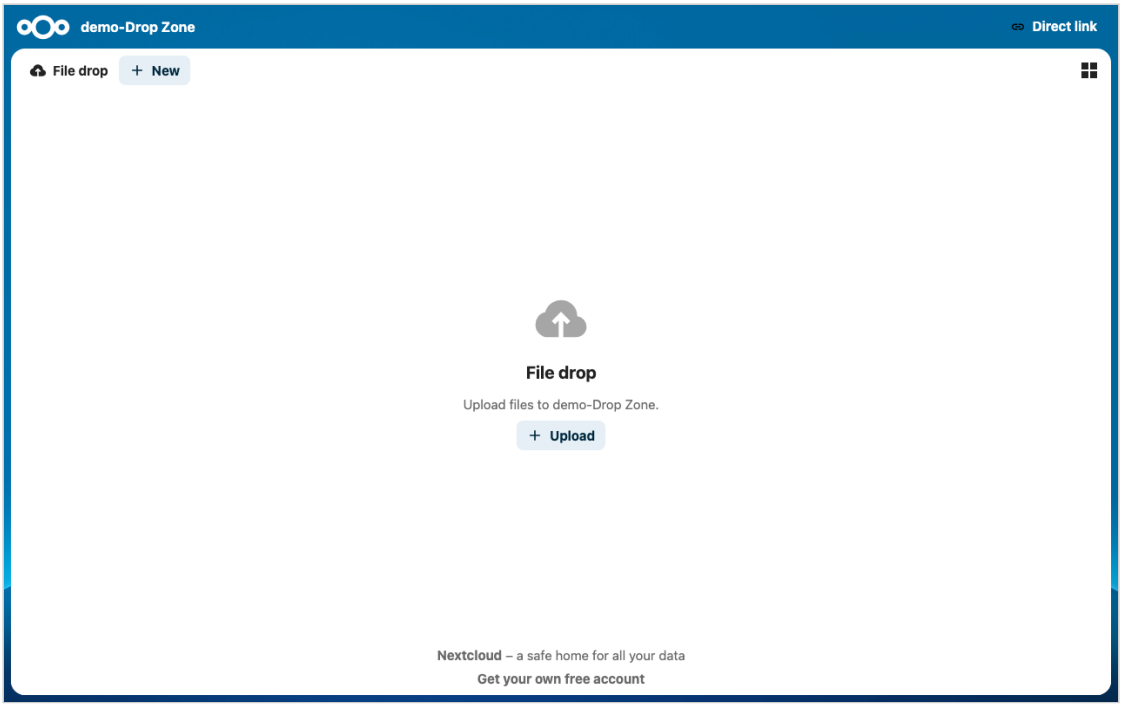


Figure 48. Uploaded

The teacher sees the file in `Drop Zone` (6.3).

8. Wikipedia offline (Kiwix)

The demo has **Wikipedia in simple English** (text, no pictures). It works without internet.

8.1 Look something up and use it in class

1. Open Kiwix from the Console (chapter 2). The **Kiwix** library shows the books available.

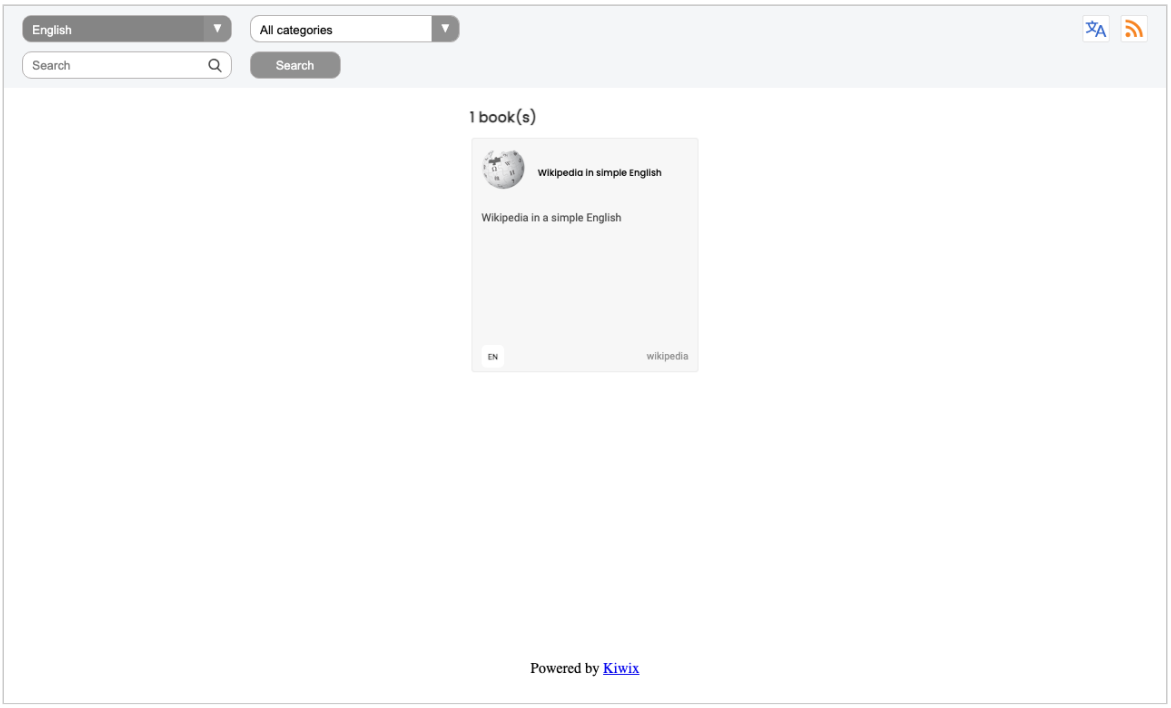


Figure 49. The Kiwix library

2. Click **Wikipedia in simple English**. Its welcome page opens.



Figure 50. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`. Pick a result.

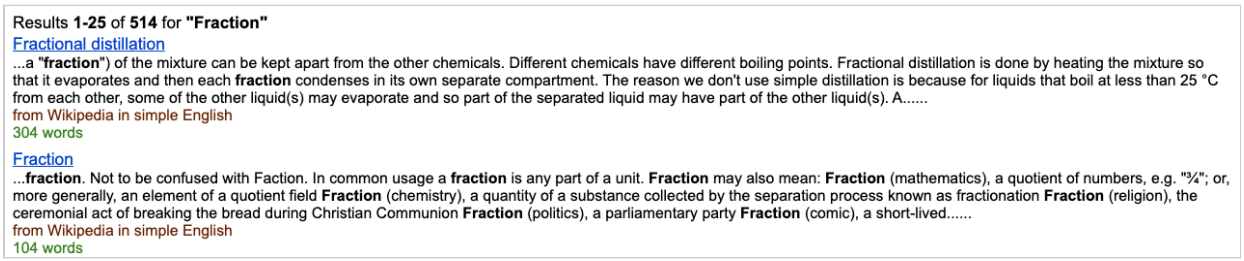


Figure 51. Search results

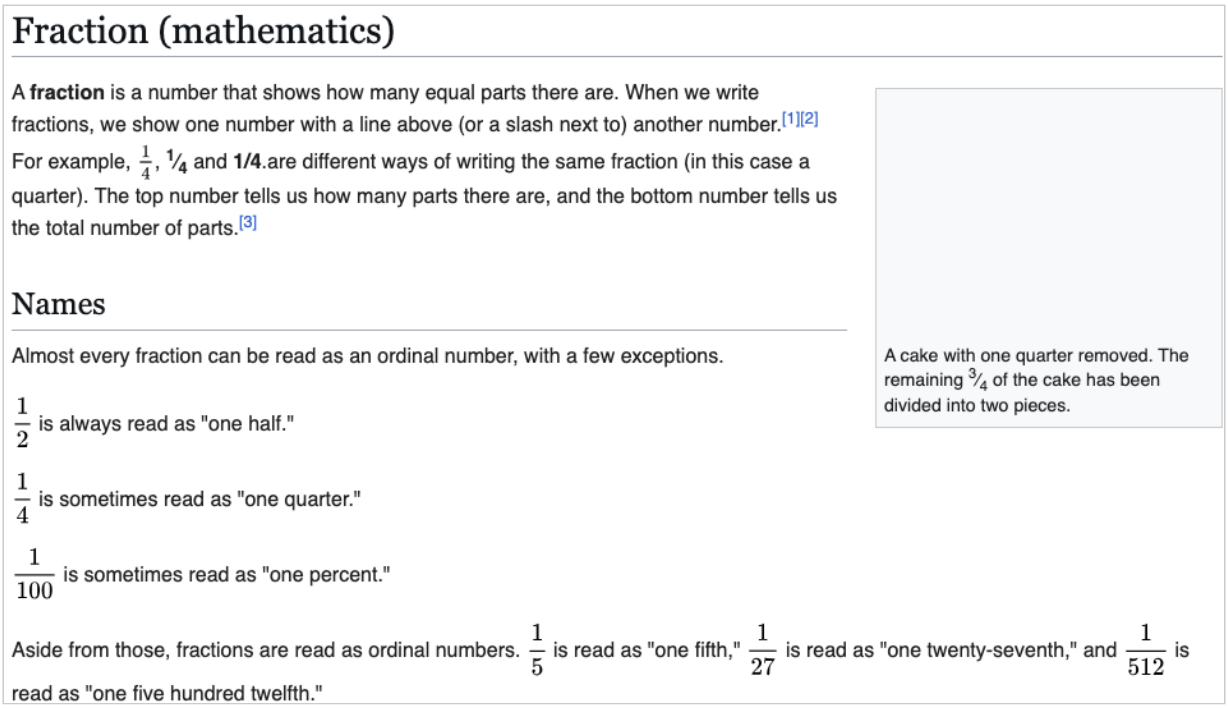
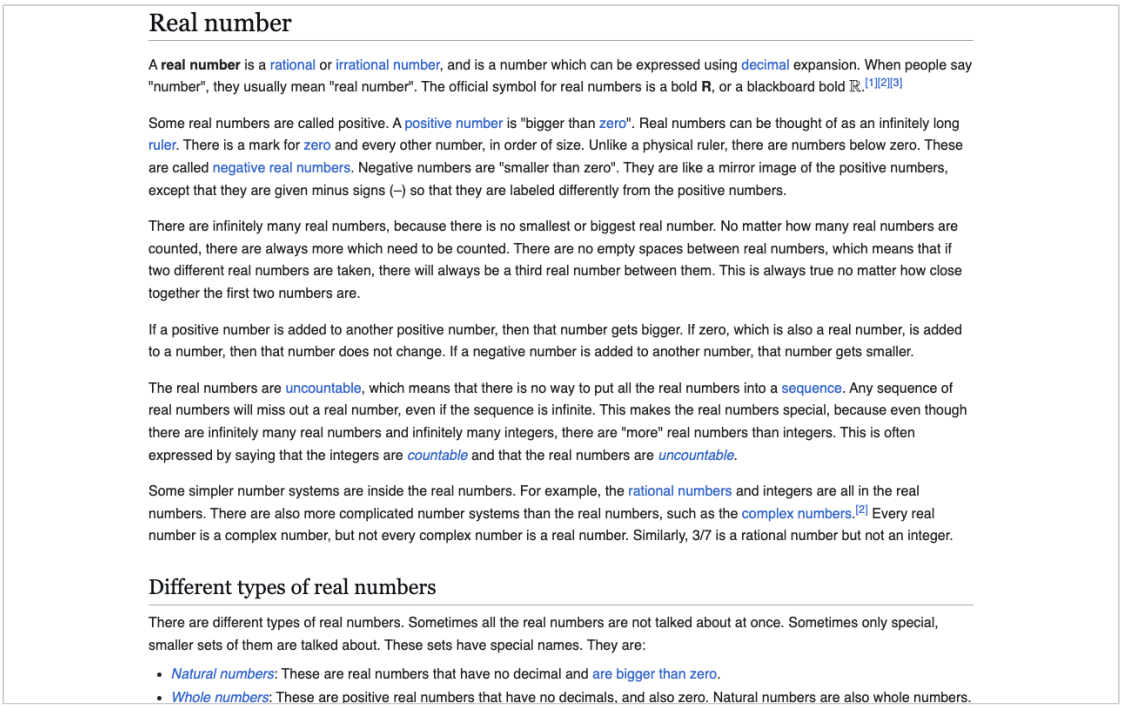


Figure 52. Fraction article

4. Follow blue links to related articles.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 53. Linked article

> **Classroom idea.** give learners 3 questions to answer from one article; they read on their own device.

9. Your own files: the Files Disk

A **Files Disk** is an **IDEA Disk** that the operator prepares in the Console (chapter 12). Its content appears in Nextcloud as the folder **School Files**. Use it to bring in material you downloaded elsewhere, for example worksheets you copied at another school.

> **Only a USB stick?** You can use a normal USB stick without a Files Disk: it appears in Nextcloud as a read-only **USB:** folder (5.5).

Good to know:

- **School Files** is only in **Nextcloud**, not in Kolibri.
- Only teachers see the **School Files** folder. Learners get only the files you share with them.
- **The Backup Disk does not save School Files.** Keep your own copy of important files.

9.1 Open School Files

1. Open Nextcloud from the Console and sign in. In **Files** you see the folder **School Files** (marked as **External storage** in the left menu).

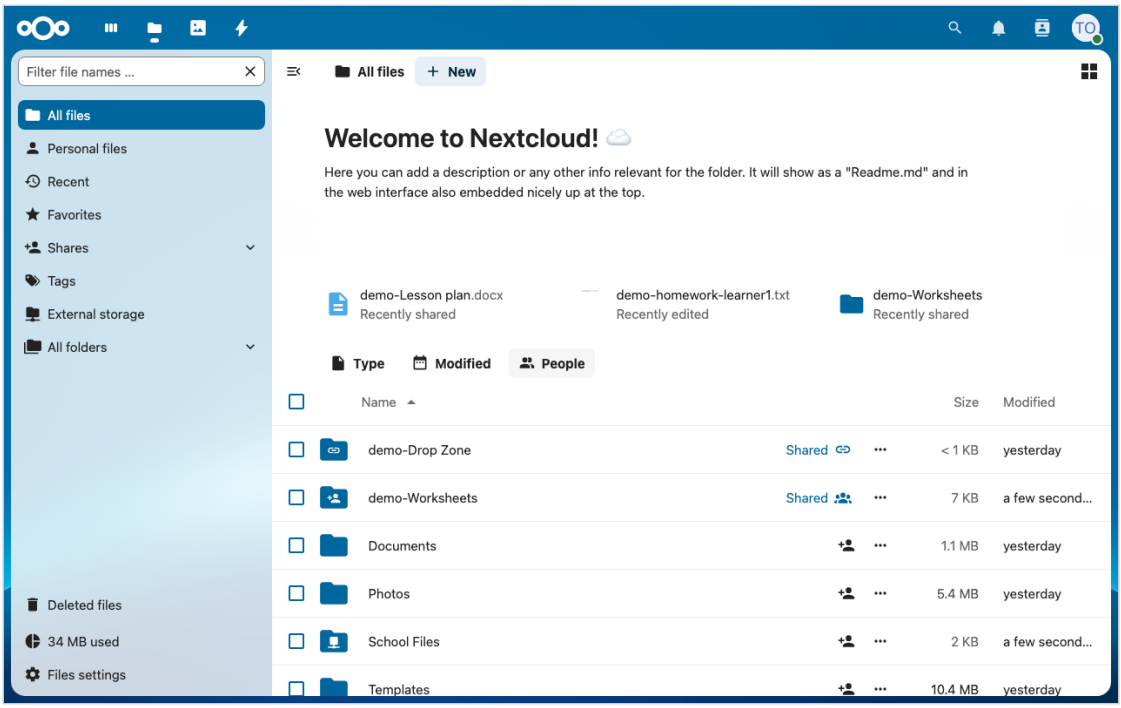


Figure 54. School Files in Nextcloud

2. Open **School Files** to see what is on the disk (here `demo-Fractions-Worksheet-1.pdf`).

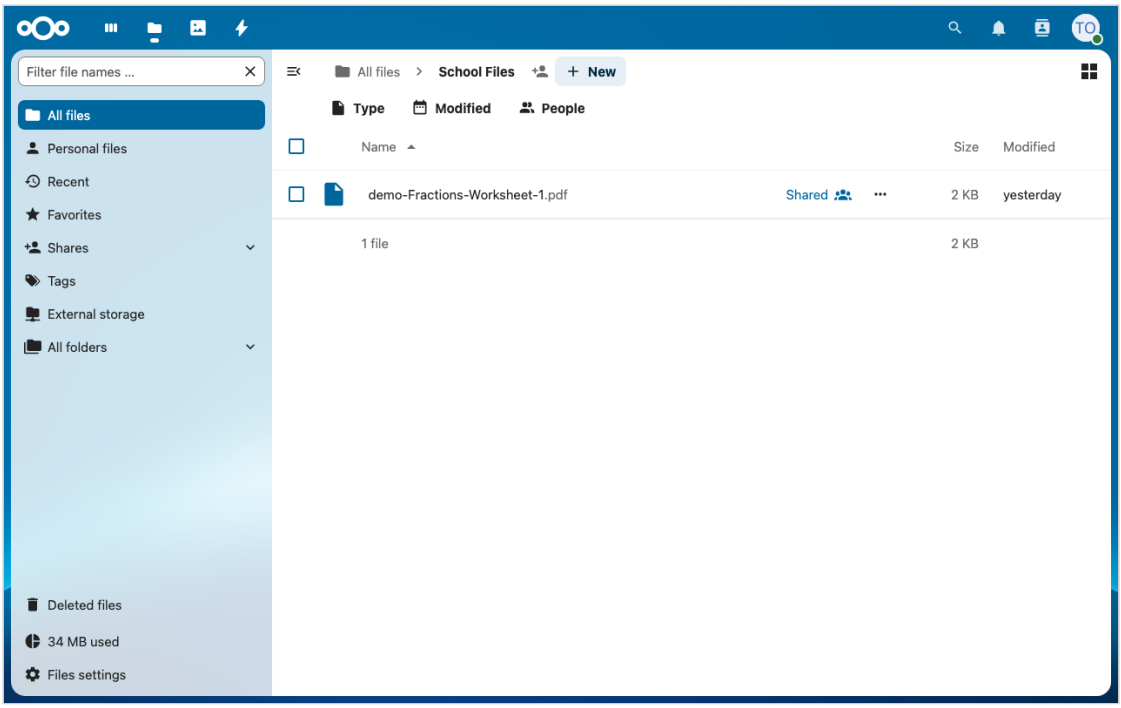


Figure 55. School Files content

9.2 Share a file from School Files with your class

1. On the file's row click **··· (Actions) > Show sharing options**, or click the share icon.
2. In **Internal shares**, type the class group name (here `demo-Grade5A`) and choose it from the list. Keep **View only**. The group then appears under **Internal shares**.

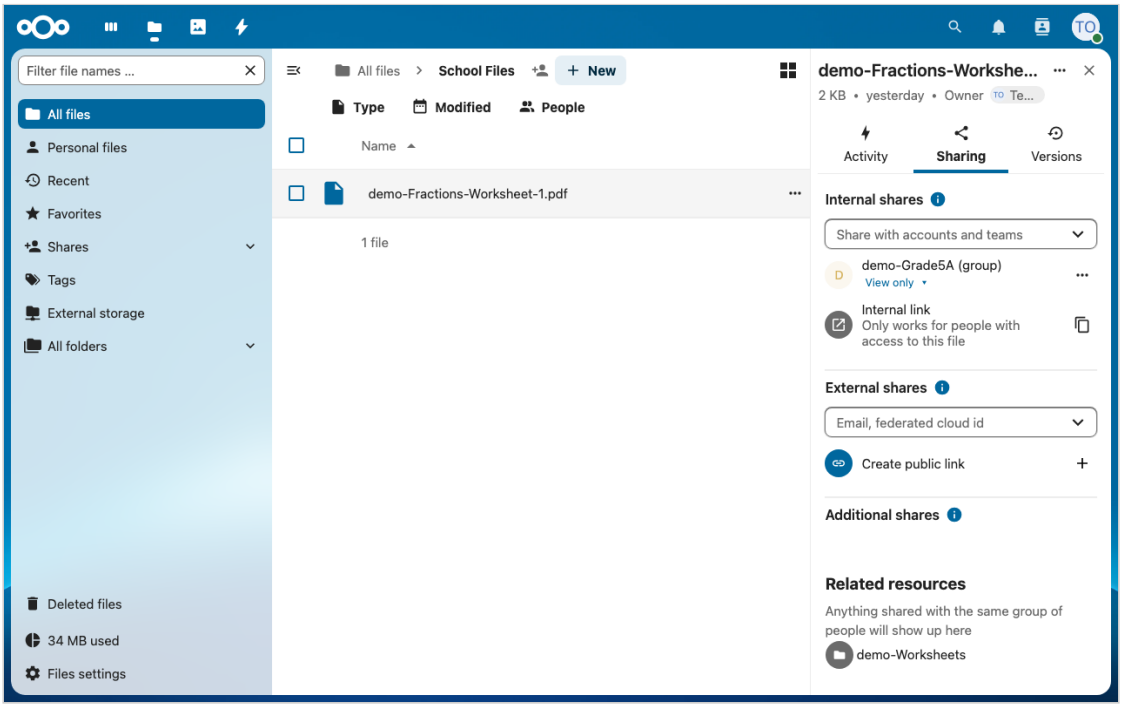


Figure 56. File in School Files shared with demo-Grade5A

3. A learner in that group signs in and finds the file in **All files** and under **Recently shared**.

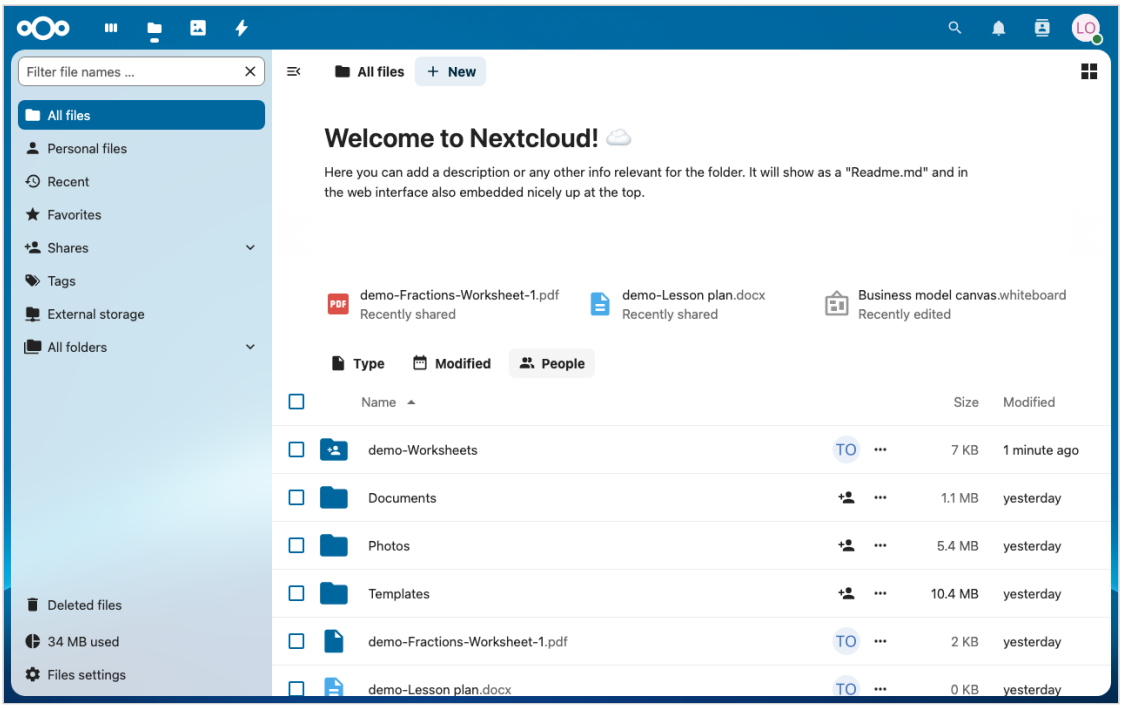


Figure 57. A learner sees the shared file

9.3 Can I put my own files into Kolibri?

Yes, with the **channel builder**: see chapter 5. It turns a folder of your files (on a USB stick, in School Files or in Nextcloud) into a Kolibri channel.

10. Keeping data safe: the Backup Disk

The operator looks after backups (chapter 12). As a teacher you should know:

- A **Backup Disk** is a separate **IDEA Disk** with the **backup** badge in the Console.
- It saves the **apps' data**: Kolibri (classes, learners, progress) and Nextcloud (accounts, and the files inside Nextcloud).
- It does **not** save the **School Files** folder (chapter 9).
- During a backup, the app stops for a short time and then starts again. Backups are planned outside lesson time.
- How often backups run (on request, each time the disk is plugged in, or on a schedule) is set by your IDEA contact.
- If the IDEA box breaks, the operator restores the latest backup onto a spare disk. Only the work done after the last backup is lost.
- Never unplug a disk while a backup is running.

Part B: For the school operator

11. The Console for the operator

This chapter is for the school operator. Do not press **Start**, **Stop** or other action buttons unless your IDEA contact asks you to.

11.1 Operator Login

1. In the Console, click the person icon (top right). The **Operator Login** panel opens with a username and a password field (the eye icon shows or hides the password).

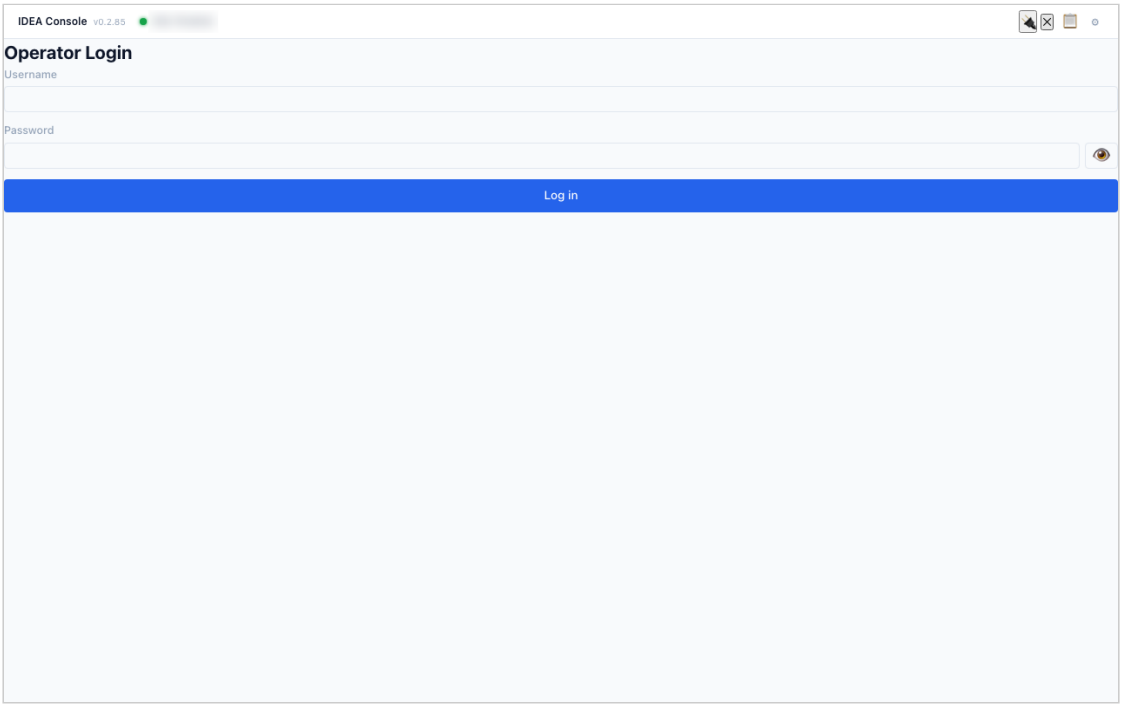


Figure 58. Operator Login

2. Type the operator username and password and click **Log in**. The **Account** panel shows your name, the role **Operator**, **Change Password**, **Manage Operators** and **Log out**.

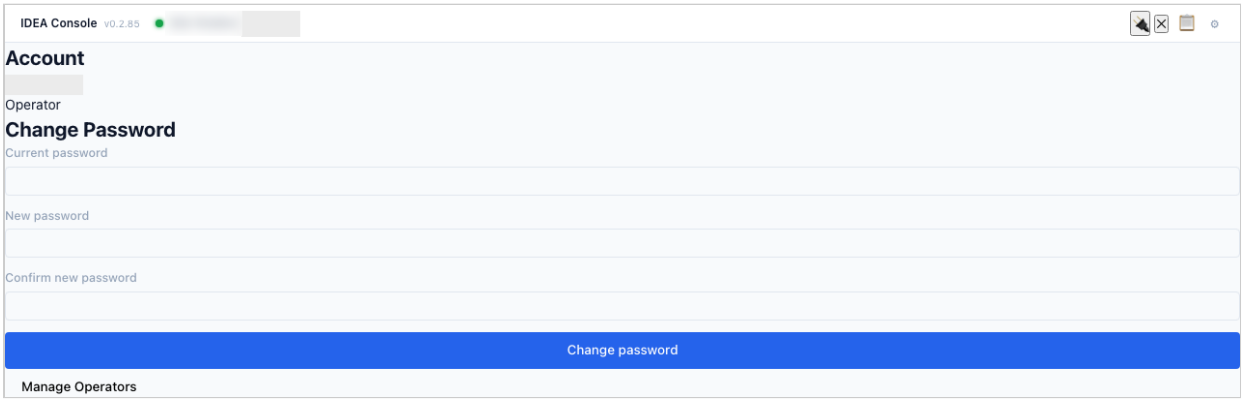


Figure 59. Account panel

3. **Change the default operator password straight away** on a new school box: **Account > Change Password**, type the current password, the new password twice, and click **Change password**. Keep the new password with the school head, not on a sticker on the box.

4. Close the panel with ×. Use **Log out** in the same panel when you are done.

11.2 The Network view: the IDEA box and its disks

1. After login the left side shows **NETWORK**: **ALL APPS**, then the IDEA box with a green dot (online) and its disks.

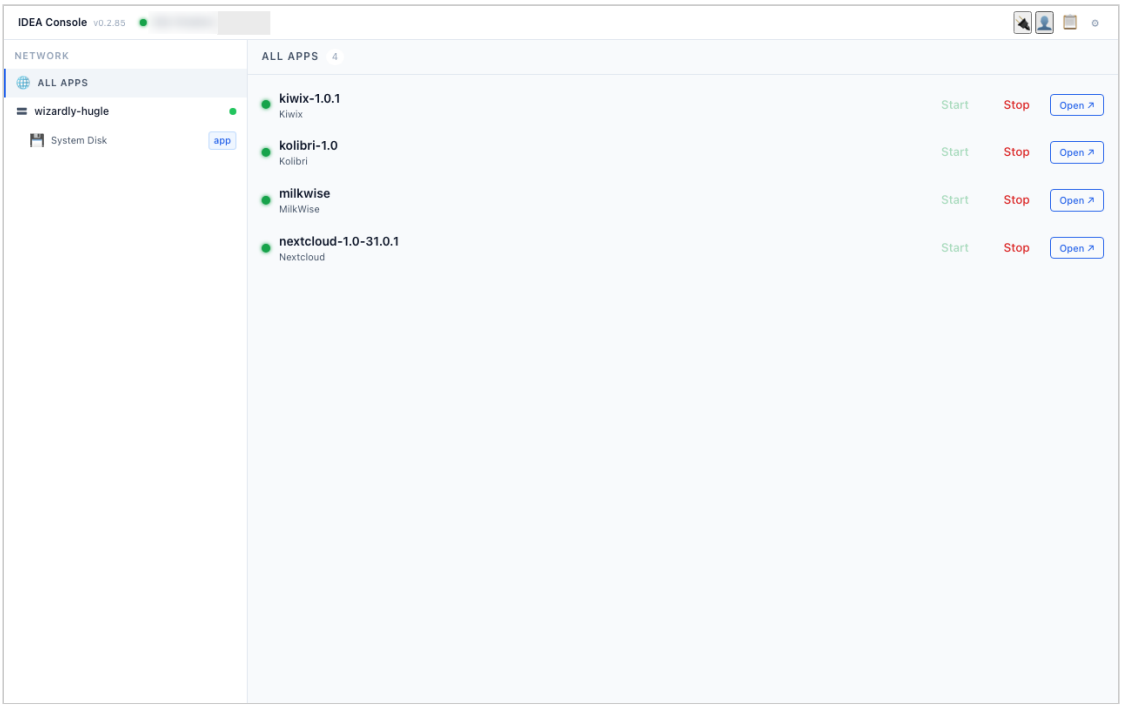


Figure 60. All apps

2. Click the IDEA box name to select it, then click a disk (here **System Disk**). Each disk shows a role badge: **app**, **backup**, **files** or **empty**. The right side lists the apps on that disk.

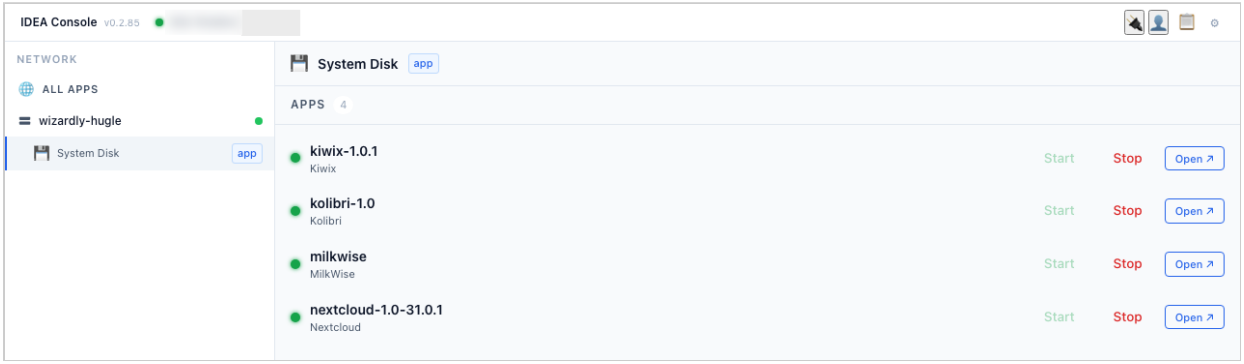


Figure 61. Disk view

11.3 App status

1. Each app row shows the app name and version, a coloured status dot, and the buttons **Start**, **Stop** and **Open ↗**.

2. Click the status dot (**Show details**) to see **STATUS** (for example **Running**), **VERSION**, **PORT**, **CREATED**, **LAST STARTED**, **LAST BACKUP** and the description.

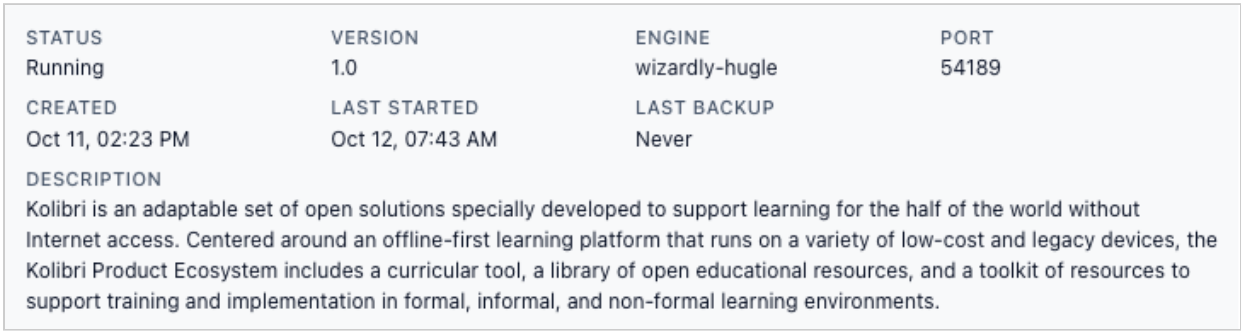


Figure 62. App details

3. To check that an app really works, click **Open ↗**: the app's page must load.
4. Only start or stop an app when your IDEA contact asks you to. Do not touch apps your IDEA contact did not mention.

11.4 Command History

The clipboard icon (top right) lists recent Console commands with ✓ (done) or ✗ (failed) and how long ago. Read it out to your IDEA contact when you report a problem.

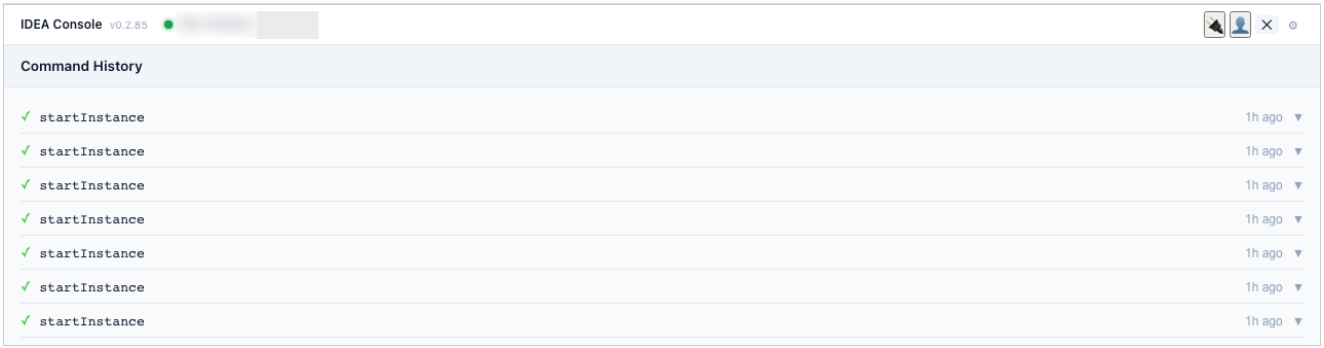


Figure 63. Command History

11.5 Buttons to leave alone

- The small ↺ next to the IDEA box name is **Reboot**: it restarts the whole IDEA box. Never press it during lessons.
- **Erase this disk…**, **Install App**, **Make this a Files Disk**, **Make this a Backup Disk**, **Restore** and **Stop**: only with your IDEA contact on the phone. The System Disk has no Erase button.

12. Disks: Files Disk and Backup Disk

This chapter is for the school operator, with the IDEA contact on the phone.

12.1 A new disk (not yet an IDEA Disk)

1. A new disk appears under the IDEA box with its model and size and an **UNFORMATTED** badge.

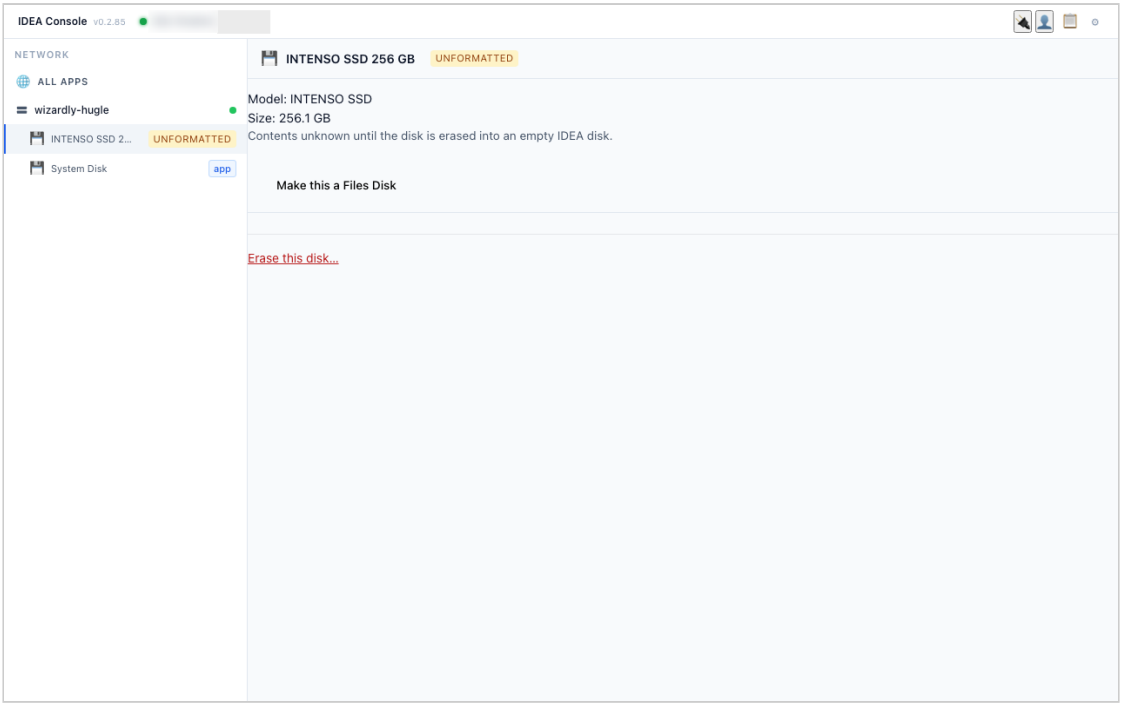


Figure 64. Unformatted disk

2. Its view shows the model, the size and two buttons: **Make this a Files Disk** and **Erase this disk…**.

USB sticks. A teacher's FAT32 or exFAT stick shows as **USB stick: (name) (read-only)**. It needs no erasing: the IDEA box only reads it, and teachers use it in Nextcloud and the channel builder (chapter 5). **Erase this disk…** is still offered, but erasing turns the stick into an IDEA Disk and deletes the teacher's files, so only do it when asked. A stick with a `kolibri` folder is built into channels automatically only if the channel builder is on this box and not stopped: if you stopped it, the stick row says so. Start it again to build.

12.2 Erase a disk

1. Click **Erase this disk…**. The confirmation shows the disk name, model, size and contents, and the red warning "This erases everything on the disk…".
2. Check the **size and contents**: several disks can have the same name (for example **IDEA Disk**). Make sure it is not the Backup Disk.

3. The box above the text field says **TYPE IDEA DISK TO CONFIRM** (the disk name in capitals). Type the name as it is written on the disk, for example `IDEA Disk`. The **Erase this disk** button stays grey until the name is right. Press **Cancel** if anything looks wrong.

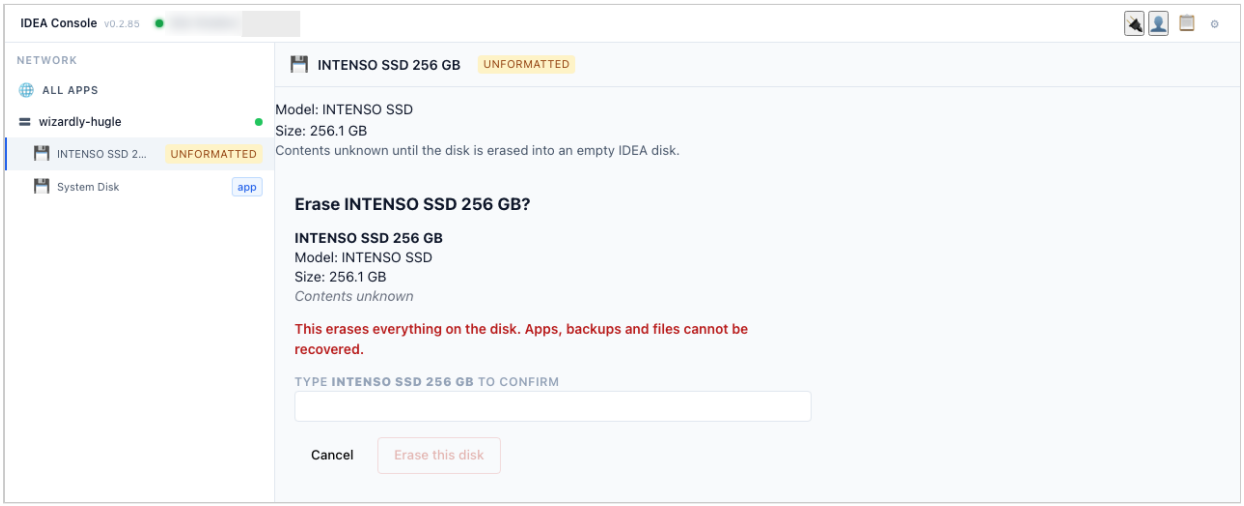


Figure 65. Erase confirmation

4. After the erase, the disk shows as an **IDEA Disk** with the **empty** badge.

> **A Files Disk that Nextcloud is using cannot be erased.** Ask your IDEA contact to stop Nextcloud first.

12.3 Make a Files Disk

1. Select the **IDEA Disk** marked **empty**. The view asks "What would you like to do with this disk?": **Make this a Files Disk**, **Make this a Backup Disk** or **Install App**.

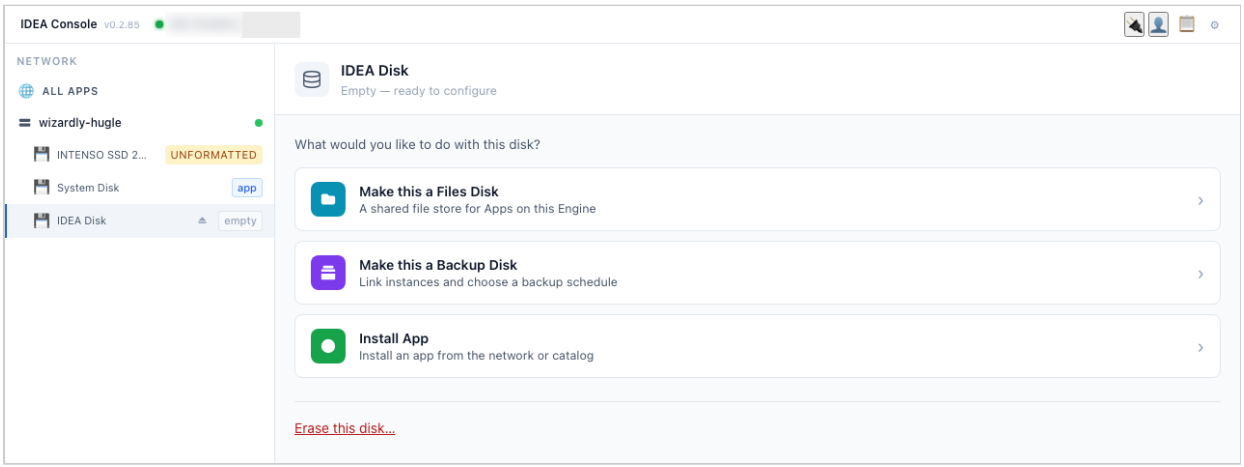


Figure 66. Empty IDEA Disk

2. Click **Make this a Files Disk**. The panel says "Nothing on the disk is erased", with **SHARE NAME** filled in as **School Files**. Click **Make this a Files Disk**.

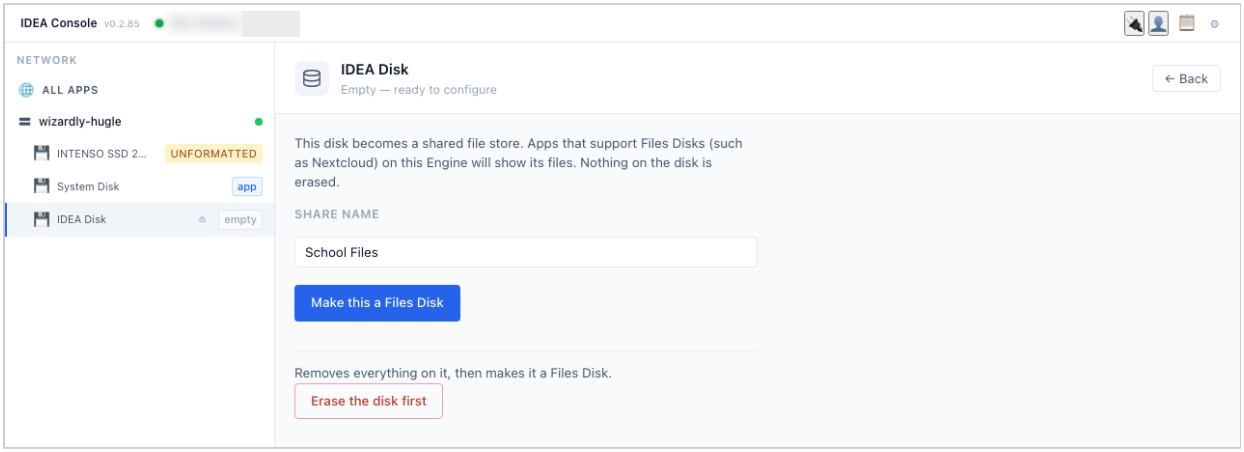


Figure 67. Make this a Files Disk

3. After a short wait the disk shows the **files** badge, "Share name: School Files", the free space, an **Eject** button and **Erase this disk…**. Once your IDEA contact has linked it, the view also says **Available in: Nextcloud**. Teachers then find it in Nextcloud (chapter 9).

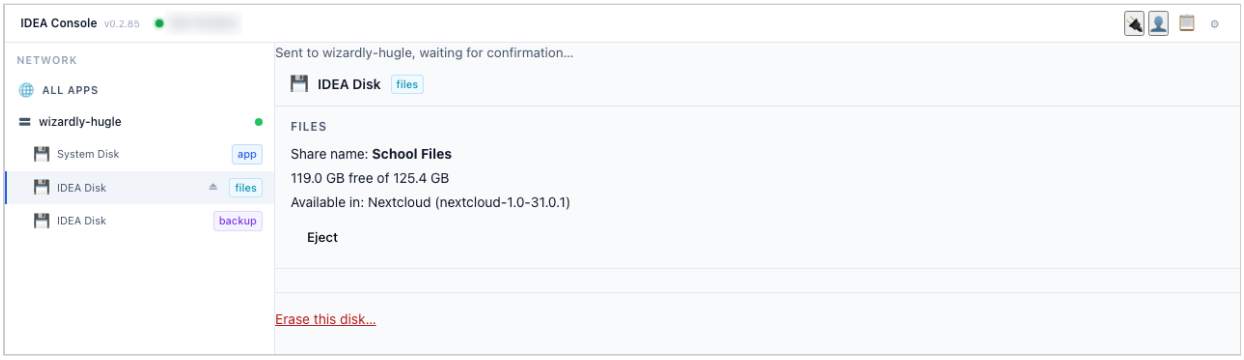


Figure 68. Files Disk available in Nextcloud

12.4 Eject a disk

An IDEA Disk has a small eject icon next to its name in the Network list and an **Eject** button in its disk view. The System Disk has none. Never eject or unplug a disk during a backup.

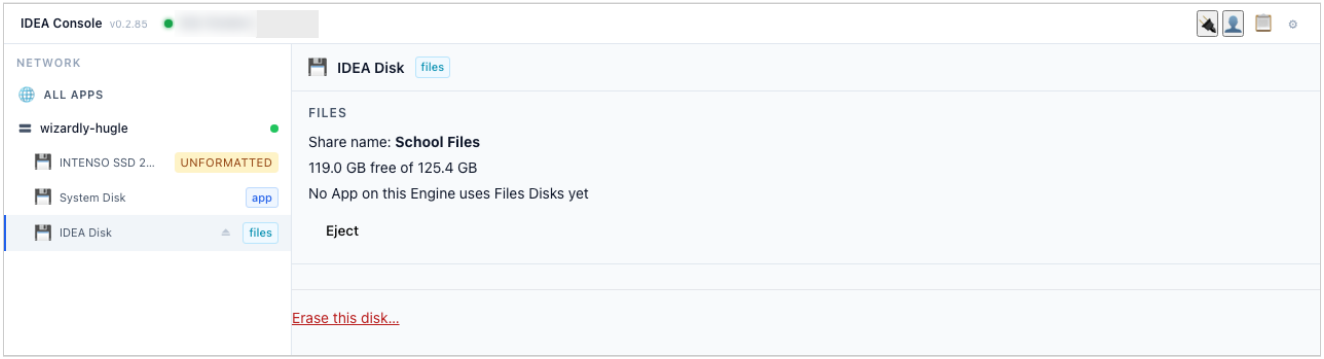


Figure 69. Eject button on a Files Disk

12.5 Make a Backup Disk

1. Select an **IDEA Disk** marked **empty** and click **Make this a Backup Disk**.

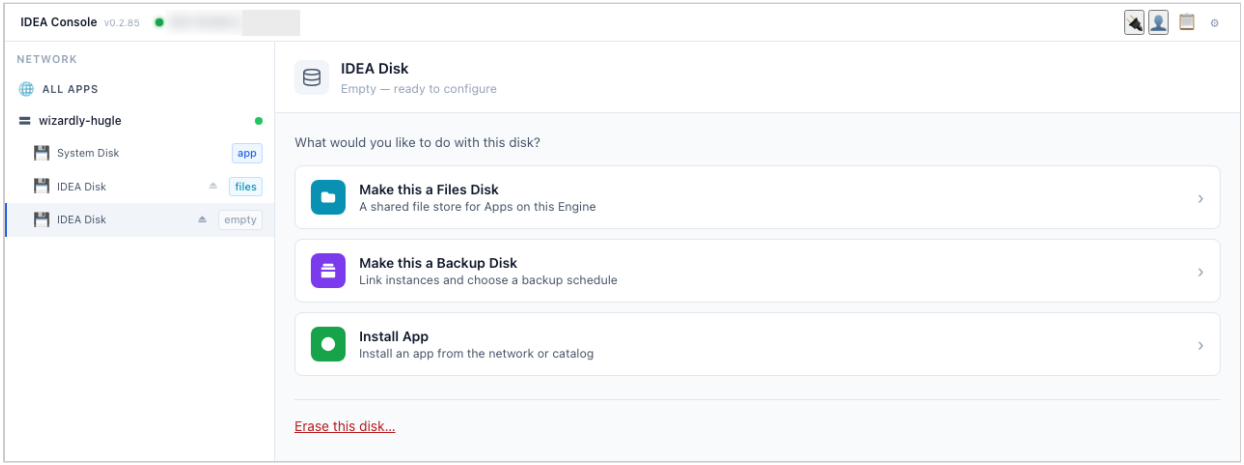


Figure 70. Empty disk options

2. Choose when backups run: **On demand** (only when you press **Back up**), **Immediate** (as soon as the disk is plugged in) or **Scheduled**. Your IDEA contact tells you which.

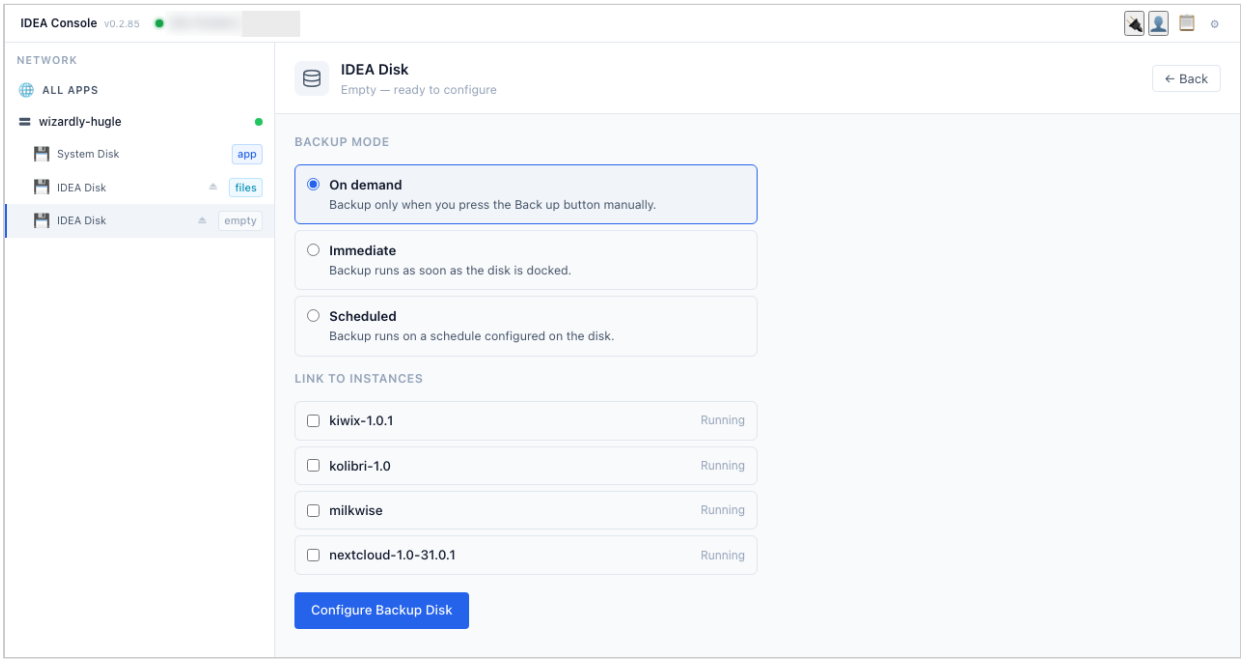


Figure 71. Backup mode

3. Under **LINK TO INSTANCES**, tick only the school apps to protect: Kolibri and Nextcloud.

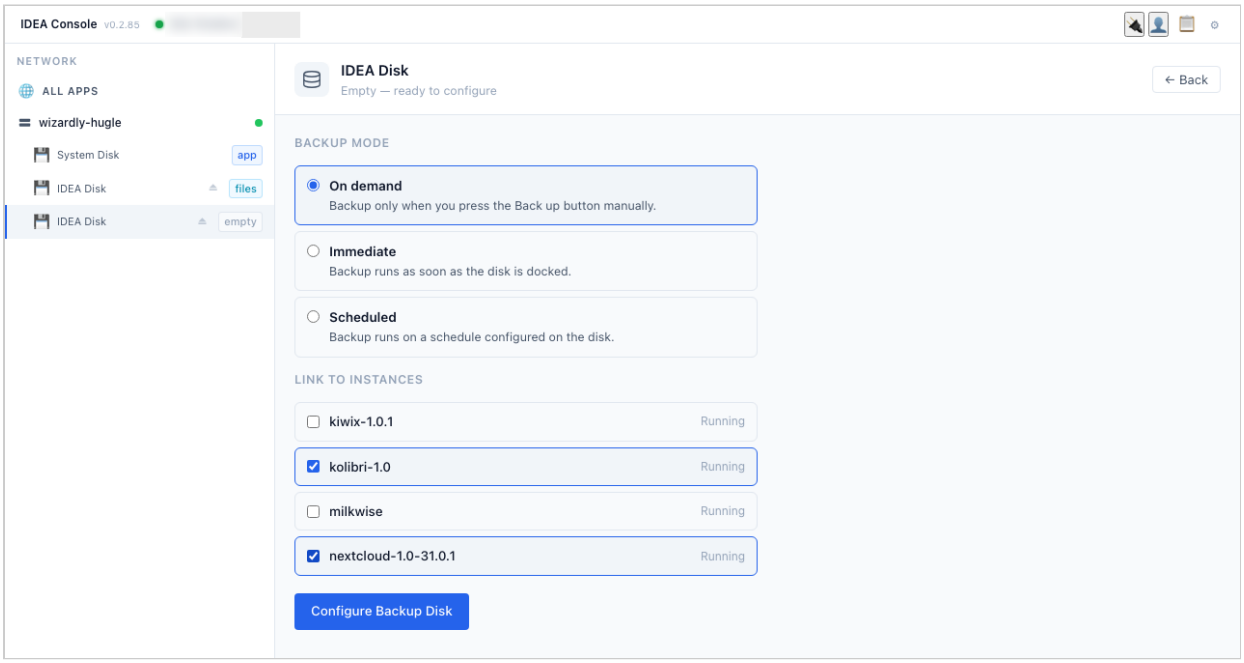


Figure 72. Linked apps

4. Click **Configure Backup Disk**. The disk gets the **backup** badge and a **Backups** list with each linked app.

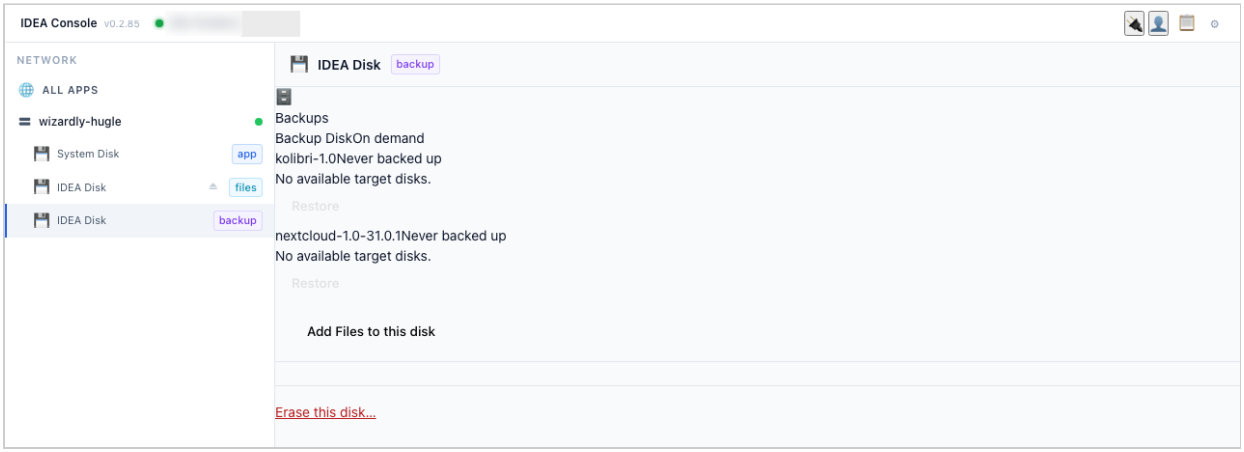


Figure 73. Backup Disk configured

12.6 Run a backup

1. In **ALL APPS**, each linked app row has a **Back up** button next to Start, Stop and Open. Click it for Kolibri. The row shows **Backing up…** with a progress bar, and the **OPERATIONS** box at the top shows the same job.

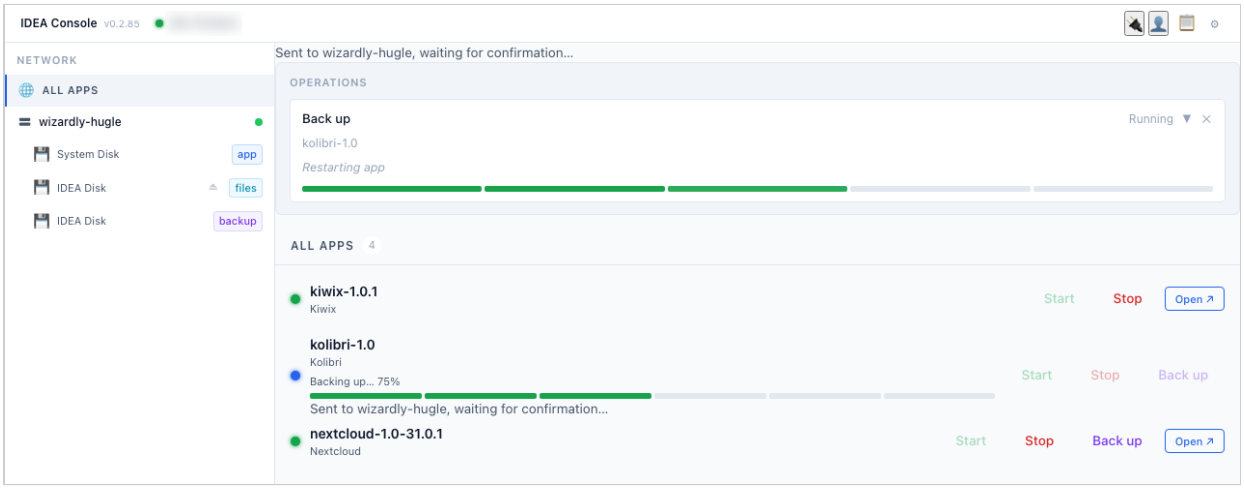


Figure 74. Back up clicked

2. The app stops for a short time while it is copied, then starts again. When the job is finished the OPERATIONS box shows **Done**; close it with ×. A backup takes a few minutes. Do this outside lesson time.
3. When it is done, the Backup Disk's **Backups** list shows **Last backup** with the date and time for each app. Repeat for Nextcloud.

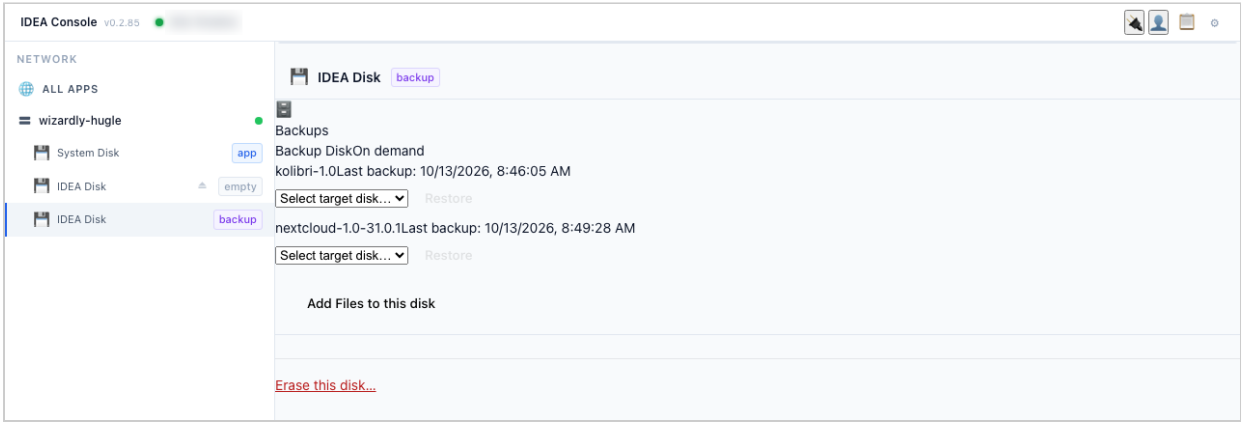


Figure 75. Last backup for both apps

12.7 Restore after a breakdown

You need a **spare disk** (an empty IDEA Disk) to restore onto.

1. Open the **Backup Disk**. Each app in the **Backups** list shows **Last backup**, a **Select target disk…** list and a **Restore** button (grey until a target is chosen).

2. In **Select target disk…** choose the spare disk. Only spare disks are offered here. Click **Restore**.

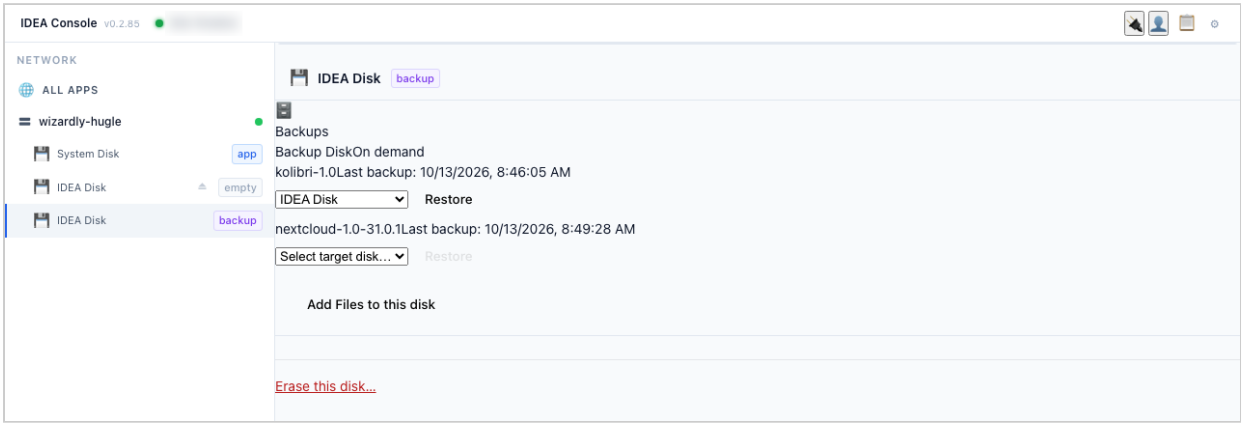


Figure 76. Target chosen

3. The Console asks: "Are you sure? This will overwrite **kolibri-1.0** on **IDEA Disk**." Click **Confirm Restore**.

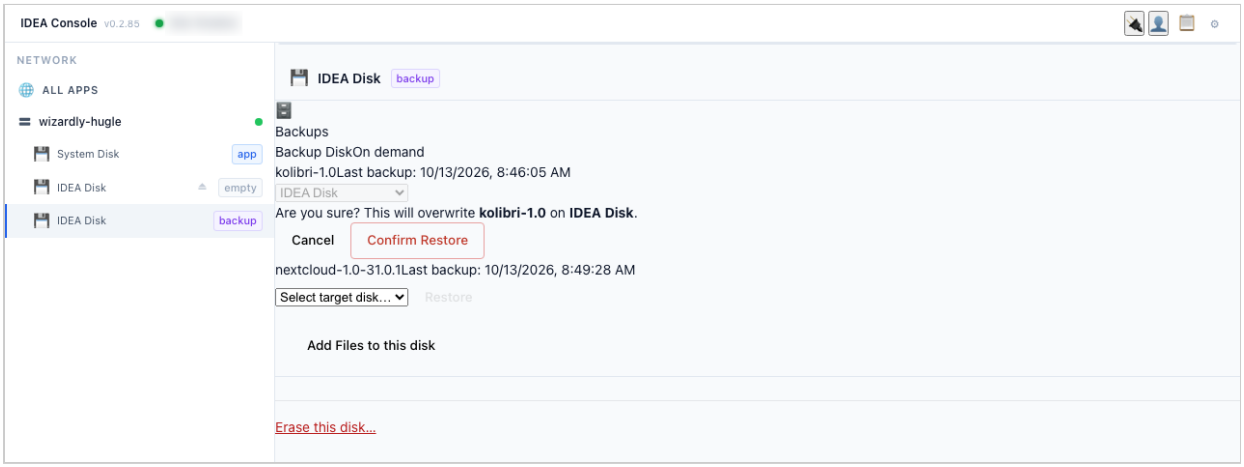


Figure 77. Confirm Restore

4. After about a minute the spare disk shows the **app** badge and lists the restored app.

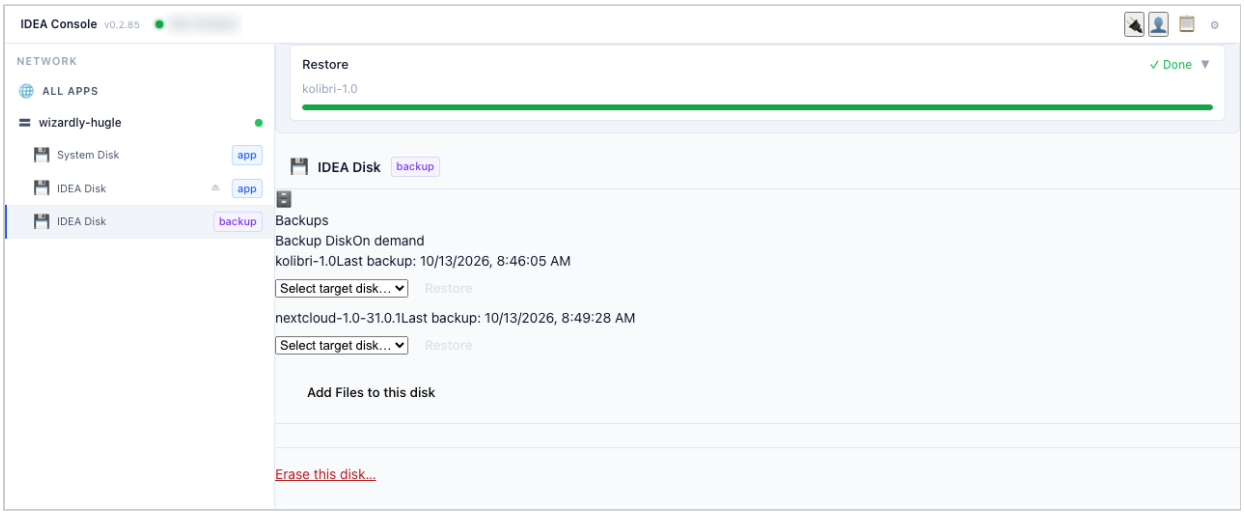


Figure 78. Restore done

5. The restore **moves** the app: it is no longer listed on its old disk, and the restored app is **stopped** (orange dot, **Start** button). Until you start it, the app cannot be opened. A restore is for a real breakdown, not for practice on a working school box.

6. Click **Start** on the restored app's row. After about 30 seconds it runs again and the row shows **Open ↗**.

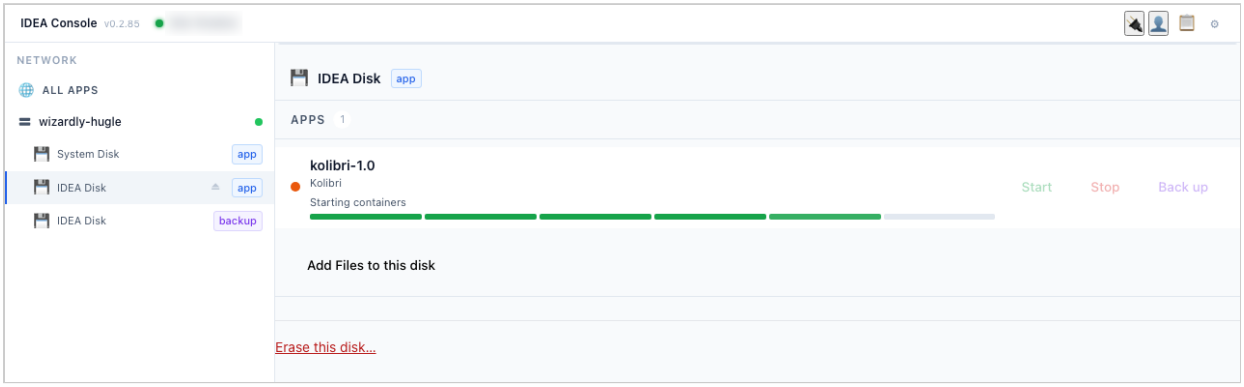


Figure 79. Start clicked

7. Open the app from its card on the **Apps** page (as in Figure 2) and sign in as the teacher (coach).

8. Check that the data is back: classes, learners, lessons, quizzes and reports, as at the last backup.

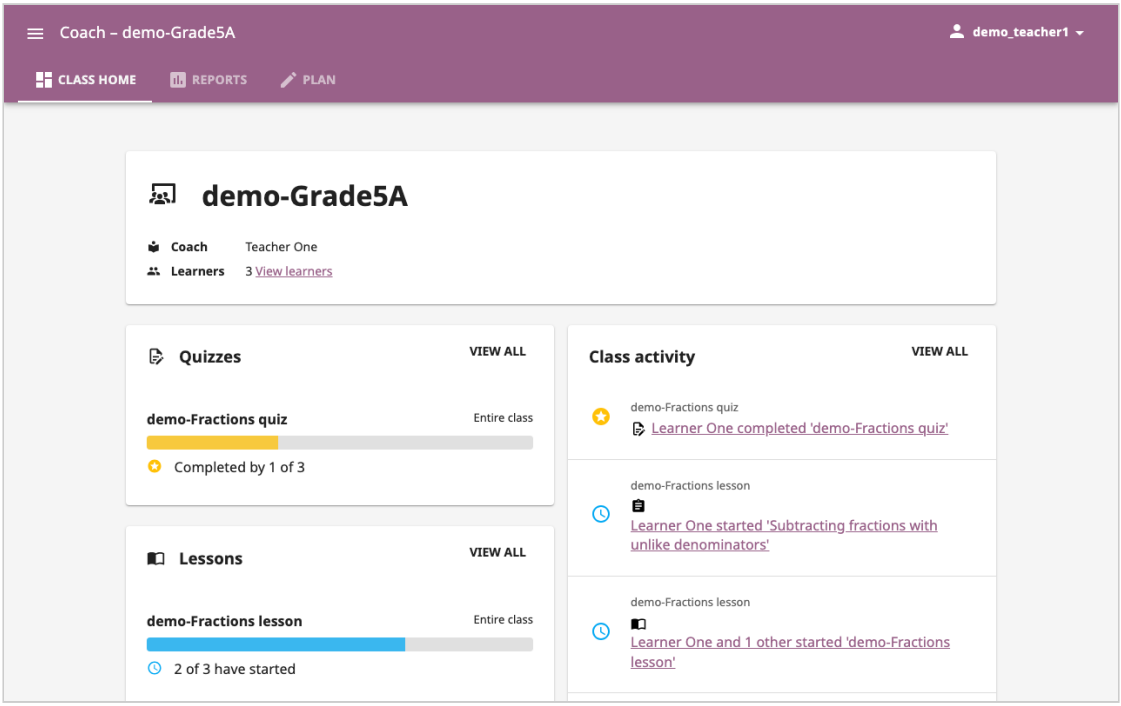


Figure 80. Class after restore

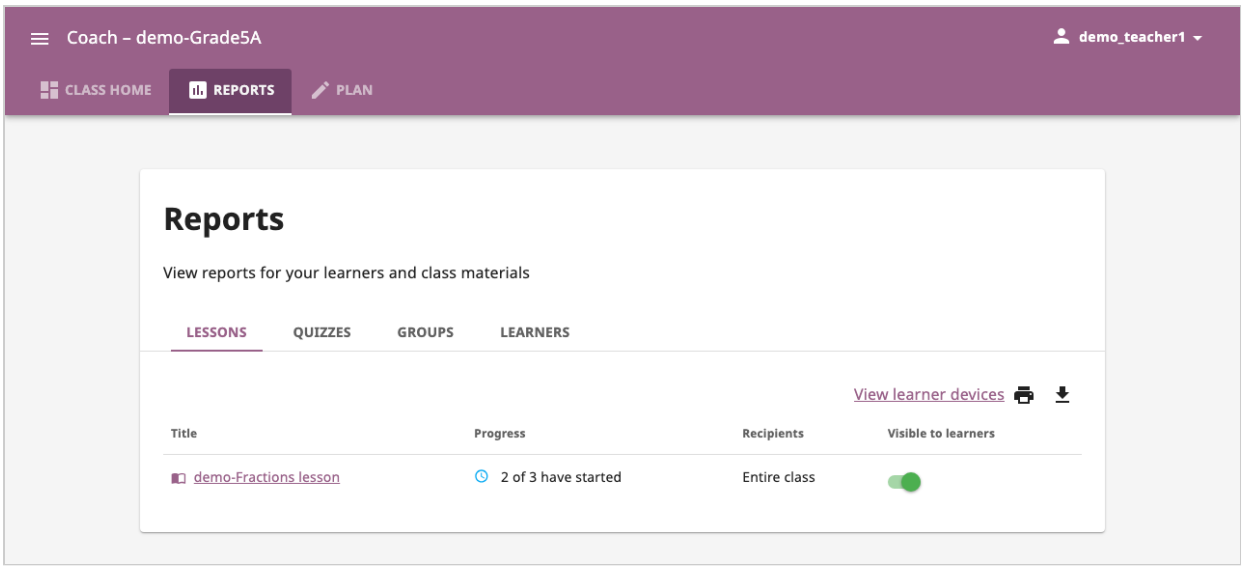


Figure 81. Reports after restore

9. Later, your IDEA contact can move the app back to the System Disk and prepare the spare disk again.

Important. Anything done after the last backup is lost. Make a new backup of the restored app as soon as it runs.

Appendix: demo accounts

These are the accounts used for the screenshots. Your IDEA contact gives you the demo passwords. Kolibri usernames use an underscore (`demo_teacher1`); Nextcloud usernames use a hyphen (`demo-teacher1`).

Username	App	Role	Used in
demo_admin	Kolibri	Kolibri admin	3.3 (shown in the enroll list)
demo_teacher1	Kolibri	Admin and coach of class demo-Grade5A	3, 5.2, 12.7
demo_learner1	Kolibri	Learner in demo-Grade5A	4
demo_learner2	Kolibri	Learner in demo-Grade5A	4
demo_learner3	Kolibri	Learner, enrolled in 3.3	3.3
demo-teacher1	Nextcloud	Teacher (groups demo-Grade5A, demo-teachers)	5.5, 6, 9
demo-teacher2	Nextcloud	Teacher (group demo-teachers)	6.2
demo-learner1	Nextcloud	Learner (group demo-Grade5A)	7, 9.2
demo-learner2	Nextcloud	Learner (group demo-Grade5A)	7.1

Demo content: Kolibri class **demo-Grade5A**, lesson **demo-Fractions lesson**, quiz **demo-Fractions quiz**; Nextcloud folders **demo-Worksheets** and **demo-Drop Zone**, document **demo-Lesson plan**.